

SIXTH INTERIM PROGRESS REPORT
ON THE PHYSICAL REALIZATION OF AN ELECTRONIC COMPUTING INSTRUMENT

by

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PREFACE

This report has been prepared under the terms of Contracts W-36-034-ORD-7481 and DA-36-034-ORD-19 between the Research and Development Service, U. S. Army Ordnance Corps and the Institute for Advanced Study. It is both a final report on the work undertaken under the former contract and an interim report under the latter one, describing the progress up to January 1, 1951.

The Institute for Advanced Study was particularly fortunate in having Dr. C. V. L. Smith, the head of the Computer Branch of the Office of Naval Research prepare this report, and the personnel of the Computer Project wish to express their gratitude to him for undertaking this onerous task and for completing it in such an excellent fashion.

Since this report was written certain changes have been made in some of the circuits described herein and certain other organs, notably the Control, have been designed. These will be discussed in a subsequent report.

John von Neumann
Project Director

Institute for Advanced Study

*4-046

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INTRODUCTION

I was fortunate in finding it possible to spend the period from July 1950 to February 1951 at the Electronic Computer Project of the Institute for Advanced Study, engaged in studying the logical organization and the engineering development of the computer. From my observations I have written the following report, which is intended to present the status of the computer development as of January 1, 1951.

As the Control Organ was not completed at that date, the reader must expect to find certain omissions in my account of the other Organs, particularly as regards those circuits necessary to interconnect them into a complete system. Apart from this, the account of the Arithmetic Organ and of the Memory are reasonably complete, while substantial portions of the Control are also described. It must be emphasized that modifications will be required in the existing circuitry as the Control is completed: these, however, will be ordinarily in matters of detail rather than in overall organization.

I have not attempted to describe the physical lay-out of the computer, as this has been done in earlier reports. The reader is referred to these for overall organization and for details of the elegant and ingenious three-dimensional organization of the individual chassis. I have concentrated my attention on the circuits, and endeavored to explain their operation in some detail.

Obviously it would have been impossible for me to write this report

without the active co-operation of the members of the staff of the Project. I thank them collectively and individually for their patience in answering my questions and helping me to clarify my understanding of their work.

CHARLES V. L. SMITH

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I. CIRCUIT ELEMENTS

Before proceeding to describe the organization of the various organs of the computer in detail, we will first give a brief discussion of some basic circuits which occur repeatedly.

First, and most important, is the familiar Eccles-Jordan circuit, the form used being illustrated in Fig. 1. This will always be referred to as a "toggle".

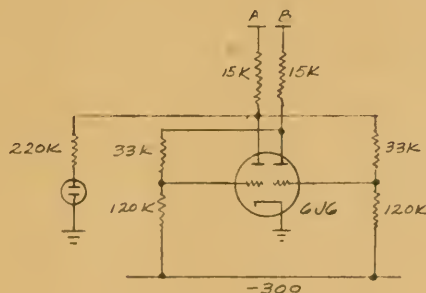


Fig. 1.

A and B designate d. c. buses: these are held at +150v except when clearing action is desired, when the appropriate voltage is dropped to +50v, this will be discussed below.

The circuit is so designed that satisfactory operation is obtained if the resistor values do not vary more than 10% from their nominal values and the tube characteristics remain within the region where the ordinates lie

between 50% and 200% of their nominal values.

If nominal values of the resistors are used, and the 6J6 characteristics as given by the manufacturer, the following electrode voltages obtain:

"on" grid	+5v
"off" grid	-40v
"on" plate	+35v
"off" plate	+100v
"on" grid current	.6ma.

A conventional representation of the toggle circuit is sometimes convenient: the one given in Fig. 2 will sometimes be used

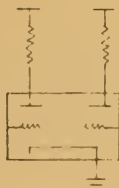


Fig. 2.

The toggle is a binary device, in the sense that at all times one section of the 6J6 is in a conducting state, while the other is not. It is merely a matter of convenience which one of the two states is chosen to represent a binary 0, the other then necessarily representing a binary 1. In order to represent visually the contents of a toggle, a neon bulb is connected from the plate of the section that is off when the toggle is holding a 1 to ground. Clearly this bulb lights up to represent 1, while it is off when the toggle holds 0. Thus in Fig. 1 we have adopted the convention that

the toggle holds a 1 when the left-hand section is non-conducting.

We now consider the "clearing" action, assuming the given convention as to the contents of the toggle. Suppose the toggle to hold a 0, and let bus A drop to +50v, while B remains at +150v. The grid voltage of the right section is merely reduced still further. Hence that section remains cut off, the grid voltage of this left section is held up, and the state of the toggle does not change. On the other hand, if bus B is dropped to +50v, while A is maintained at +150v, the grid voltage of the left-hand section is reduced below cut-off, and the toggle is changed to its other state. Similarly, if a 1 is held, dropping B to +50v while holding A at +150v has no effect, while dropping A to +50v while holding B at +150v changes the state of the toggle.

Putting this differently, the operation of dropping A to +50v while holding B at +150v guarantees that the toggle will subsequently hold a 0, while holding A at +150v and dropping B to +50v guarantees that it will subsequently hold a 1. Thus these operations are respectively referred to as "clearing to 0" and "clearing to 1". The clearing action is produced by pulsing the appropriate bus. With the present toggles, it is necessary that the bus voltage should remain below +70v for approximately one microsecond to produce the desired clearing.

In a few places in the computer, it is desired to use a toggle which requires a shorter time to change from one state to the other. In these applications a so-called "super-toggle" is used, which changes its state in half the time required by the ordinary toggle. The circuit is given in Fig. 3.

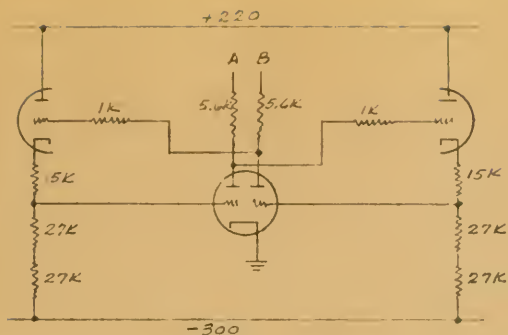


Fig. 3.

It has been shown how a toggle may be set in a desired state, or "cleared". It is also necessary to be able to set a toggle into the same state as another toggle: that is, given toggles A and B, it is necessary to provide a mechanism which guarantees that, after some definite time, the contents of A are identical with those of B. This is readily accomplished by means of a gate circuit as shown in Fig. 4.

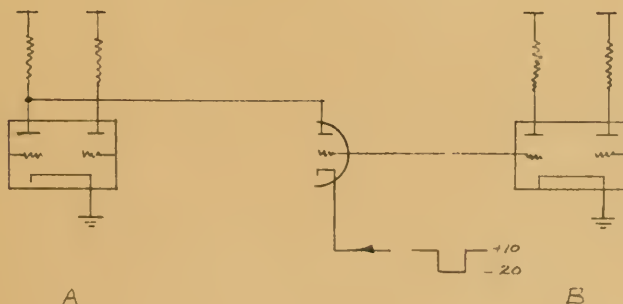


Fig. 4.

The tube used here is one section of a 6J6, with cathode normally maintained at +10v. When it is desired to transfer information from B to A (note that the given circuit permits a transfer to be made only in this direction), the cathode of the gate is dropped to -20v. The gate cannot conduct while its cathode is held at +10v, as its grid cannot be above about +.5v. Let us use our earlier convention as to the representation of binary 0's and 1's by states of the toggle. If B holds a 1, clearly nothing can happen when the gating signal arrives at the cathode of the gate, as this tube must remain cut off. If, on the other hand, B holds a 0, conduction takes place when the cathode of the gate falls to -20v, and plate current, flowing through the plate load resistor of the left section of A, will cause A to assume the state representing 0. The gate thus guarantees that if B holds a 0 at the time the gating signal is applied, A will be caused to hold a 0. Suppose now that before the arrival of the gating signal, A is cleared to 1. Then if B holds a 0, the gate causes A to assume the 0 state, while if B holds a 1, the gate has no effect, and A continues to hold a 1. Thus the sequence of signals "clear A to 1" and "open gate into A", results in A holding the same information as B.

We also note that if the gating signal were made sufficiently negative, enough grid current could be drawn by the gate to cause a change of state in B. This effect is not used in the present design, but could conceivably be useful.

The gate circuit as described is a coincidence device, equivalent to the formation of a logical product. Other methods are used in parts of the machine; for example, consider the circuit of Fig. 5, where the signals applied to the grids assume the levels of +10v or -10v. This is a cathode

follower, and the cathode will follow the highest grid. Thus the cathode voltage will assume the lower level if and only if both grid voltages assume their lower levels.

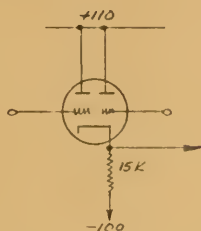


Fig. 5.

Another gate circuit which will be found in the computer makes use of two diodes connected as in Fig. 6. (see for example DWG 1325).

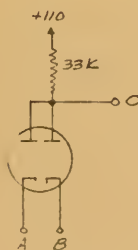


Fig. 6.

In one particular case A assumes one of two levels (+10v, -20v) and B also one of two levels (+10v, -10v). The voltage at C will always be slightly above that of the lower cathode, and hence will assume a high value (slightly above +10v) only if both A and B simultaneously attain their higher values.

Usually we think of one input to a gate as an enabling signal, which places the gate in a condition to pass on the other input. A convenient symbol for a gate is given in Fig. 7,

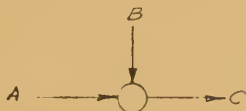


Fig. 7.

Where A and B are the inputs and C the output; the enabling signal is B. Sometimes it is convenient to use a solid arrowhead for a static voltage, and reserve the open arrowhead for a voltage pulse.

Cathode followers are found in considerable numbers. These are used, for example, to drive a number of gates of the type illustrated in Fig. 4, and in other places where it is necessary to supply a considerable amount of current from a low impedance source. The cathode followers are ordinarily operated in that region of the characteristics, where sufficient plate current is drawn to cause the cathode voltage to rise slightly above the grid voltage, so that no grid current flows. For purposes of ordinary analysis, we shall commonly assume that the cathode voltage is exactly the same as that of the grid, the cathode rise being neglected. This will ordinarily not lead to any difficulties.

II. THE ARITHMETIC ORGAN

THE REGISTERS

A Register is essentially merely a device which serves as the repository of a single word of machine language. The term is usually used to designate a storage device the contents of which are directly available with very slight delay. Thus in a machine whose language consists of forty bit words, a set of forty toggles forms a convenient Register. However, mere storage of information does not suffice, as the Registers which are to serve as the Accumulator and the AR and SR Registers of the machine must be capable of other functions. These devices must be able to transmit and receive information, and to perform shifts (at least for certain purposes) of the information to the left or to the right.

The transmission of information into or out of a Register consisting of toggles is readily accomplished by the gating circuits that have already been discussed, while the simple means by which toggles can be cleared to 1 or to 0 have also been described.

In performing the shifting operation, it is desired that the information should never exist as merely transitory electric or magnetic fields, but should rather at all times exist in static form. An effective method of accomplishing this is as follows: two rows of toggles are provided in the Registers of the machine the two rows (upper and lower) in Register RI are called R^1 and R_1 , and similarly for the other two Registers-and gates are arranged whereby the information held in a toggle of the lower row is transmitted to the left or right in two steps: first, up to the corresponding toggle of the upper row, then second, from this toggle down to the left

or to the right. Fig. 8 shows how this is done: it represents the columns $n-1$, n , and $n+1$ of the Register.

Two gates lead up from each toggle of the lower row to the toggle in the corresponding position of the upper row, while from each toggle in the upper row a gate leads down to the toggles in adjacent columns of the lower row. The gate G(reen) can transmit a 0 up, while Y(ellow) can transmit a 1. Hence to assure the transmission of the number in the lower row of toggles to the upper row, two alternative procedures are at our disposal: the top row may first be cleared to 0 and then the Yellow gates opened, or the top row may be first cleared to 1 and then the Green gates opened.

To shift the number now standing in the top row of toggles back down to the lower row, in such a way that the resulting contents of the lower row will be the number that originally resided in that row, but shifted one place to the right or to the left, is the function of the gates R(ed) and B(lack). For the right shift, the Black gates alone are available: from the diagram, it is clear that these can transmit only 1's, so that it is necessary first to clear the lower row to 0's before opening the Black gates. For the left shift, the Red gates can transmit only 0's, so it is necessary first to clear the lower row to 1's.

It was shown in the discussion of gates, that it is quite possible to cause a gate tube to clear the toggle from which it transmits information. Thus the Green gates could be used to clear the toggles of the bottom row to 1's, while the Yellow gates could be used to clear these toggles to 0's. Similarly, the Red gates could be used to clear the toggles of the upper row to 1's, while the Black gates could be used to clear these toggles to 0's. Although this property of the gate tubes is not exploited in the

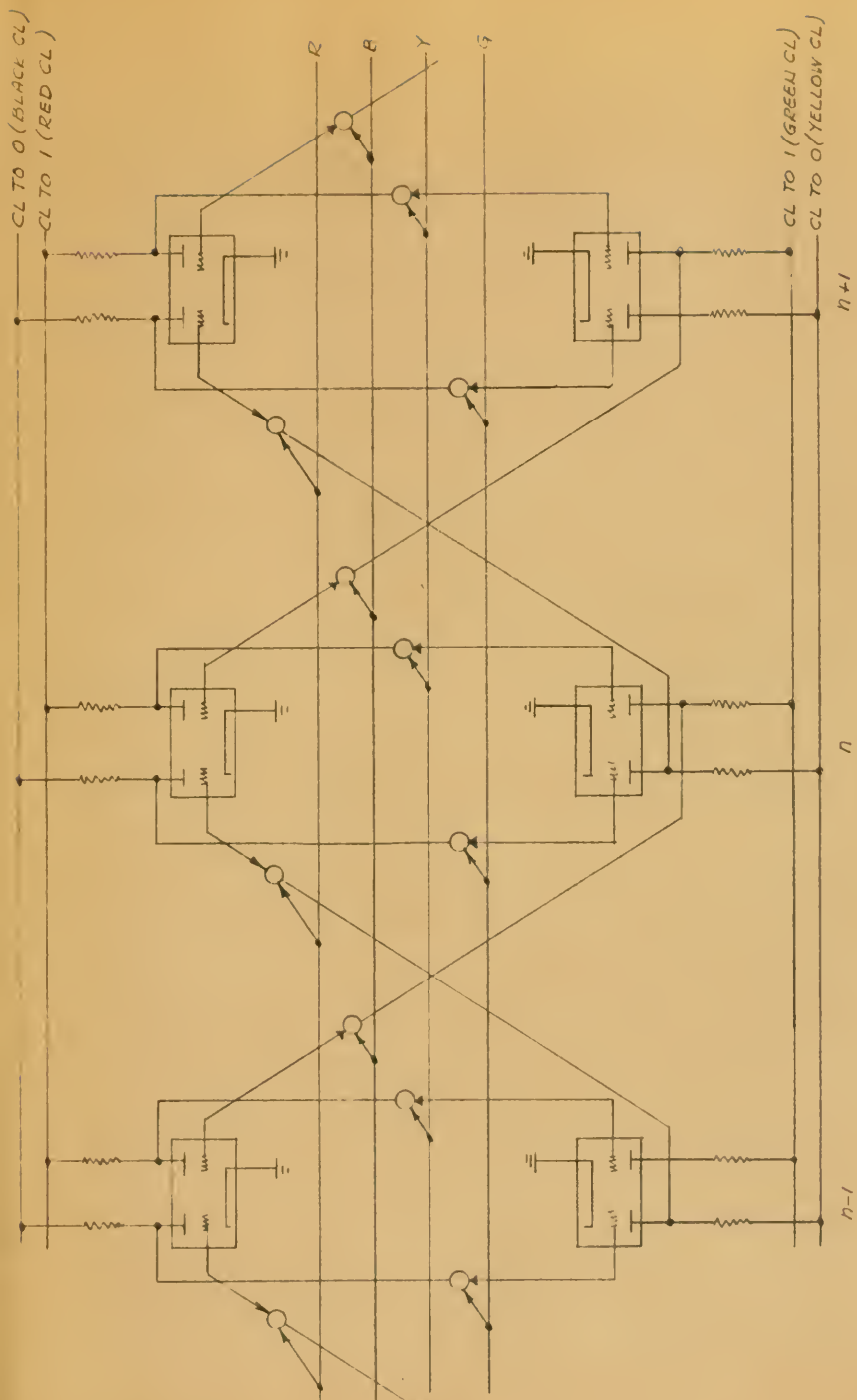


Figure 8

present machine, it nevertheless provides a convenient manner of designating the various clear operations. Thus a clear will be described by the color used to designate that gate which would be used to perform the clear. The following table shows how the color code is used to describe the various shift and clear operations:

Red:	Shift left down zero Clear top to one.
Green:	Shift up zero Clear bottom to one.
Yellow:	Shift up one Clear bottom to zero.
Black:	Shift right down one Clear top to zero.

In this terminology, the process of shifting right one place consists of the following operations: Black clear, Yellow gate, Yellow clear, Black gate. The left shift is described by: Black clear, Yellow gate, Green clear, Red gate. In each case, of course, there is an alternative process available which makes use of the Green shift up.

The structure of a forty stage Shifting Register follows directly from the diagram of the three stages which has been discussed. The actual Shifting Registers in the machine all differ in certain details from the one described, which may be thought of as an archetype. It is clear that means must be provided by which information can be inserted in or extracted from the Registers. This is done by changing certain of the connections to the gates. Registers RI and RII remain Shifting Registers with these special connections. RIII, on the other hand, should not be thought of as a Shifting Register at all. The two rows of toggles in RIII actually form two independent Registers, which form part of the means of communication between the

Williams Memory and the rest of the machine. Also a small amount of additional equipment is associated with the various Registers. In the interests of clarity, each of the three Registers will be briefly discussed and the interconnections with the rest of the machine indicated.

All three Registers consist physically of four chassis, each of which contains four rows of tubes, laid out as follows:



Fig. 9.

All tubes in the chassis are 6J6's except for those enclosed by the dotted rectangle, which are 5687's. In the first and fourth rows, the first five and the last five tubes are toggles. In the second and third rows, the first and last five tubes are gates - actually each tube is a pair of gates. The tubes enclosed by the dotted rectangle are cathode followers used as gate drivers. The two in any row are connected in parallel, and serve to drive five 6J6 gate tubes.

In each chassis, suppose the columns of tubes, exclusive of those within the dotted rectangle, to be numbered n , $n+1$, ..., $n+9$, where $n = 0, 10, 20, 30$ for the four chassis, respectively. Since numbers are represented in the machine with decimal point at the left, it is convenient to refer to any stage of a Register as 2^{-n} , 2^0 being the "sign digit".

RI and RII are identical except for certain circuits exterior to the Register Chassis: these will be described later. A schematic is given in DWG No. 1322. It has already been observed that RIII is not a Shifting Register at all. It actually consists of two Registers and associated gate tubes by means of which information can be inserted in R^3 and R_3 : for a schematic, see DWG No. 1323, in which the top row of tubes shown are the Complement Gates used to read out to the Adder, and are not actually mounted on the Register Chassis. Despite these differences, it is convenient to use the same nomenclature for the gate tubes of all three Registers. In all cases, the row of tubes directly below the toggles of R^k are alternately from the left end of the chassis Red and Black gates, while those of the row below are alternately Yellow and Green gates. Each gate tube, being a 6J6, contains two gates. (Note that the two center columns in each schematic are occupied by Gate Drivers. Note also that in each top row of toggles the convention is that 0 is signified by conduction of the left section, while in the bottom row the opposite is true.) As to the physical arrangement of the gates, the Red gate from column n to column $n-1$ is in column n , while the Black gate to column n from column $n-1$ is in column $n-1$, n signifying as before the first column in any Register Chassis. Furthermore, the Yellow gate between the toggles of column $n-1$ is physically located in column n . Thus in a complete forty stage Register, it is necessary to provide a Yellow gate for the 39th column, while half of the Yellow and Red gate tubes of column 0, and half of the Black gate tube of the 39th column are not actually used within the 40 Register stages, and hence are free for other uses: we shall see what advantages are taken of these circumstances when we examine the "end-around" circuits.

Now let us examine the actual uses of the gate tubes. In RI and RII the Red, Black, and Yellow gates are connected and used exactly as in the archetypal Shifting Register already described. The Green gates are, however, used to transmit information into R^1 and R^2 from other sources: into R^1 from the Digit Resolver, and into R^2 from R^3 . In RIII the case is quite different, all gates being used for communication between the Register toggles and other parts of the machine, according to the following scheme:

- (a) Red: into R^3 from the Discriminator in the Williams Memory.
- (b) Black: into R^3 from R_2 .
- (c) Yellow: into R_3 from the Discriminator in the Williams Memory.
- (d) Green: to Dispatch Counter from R_3 .

We have seen how information is inserted in the Register toggles. Outputs are taken as follows:

- (a) RI: from right-hand grids of R_1 toggles to Adder;
- (b) RII: from right-hand grids of R_2 toggles to RIII Black gates;
- (c) RIII: (i) from grids of R^3 toggles to Adder via the Complement gates;
- (ii) from left-hand grids of R^3 toggles to RII Green gates;
- (iii) from RIII Green gates to Dispatch Counter (already noted in the preceding paragraph).

RI has certain other circuits associated with it. One of these is an extra column to the left of the 0-th, in which the digit held in the 0-th can be

held in the case of a left shift, instead of being lost. The other is the "end-around" arrangement, which, in the case of a right shift, transfers the digit in the 39th column into the 0-th column of R_2 .

The extra column is as shown in the diagram. The tube used to provide the Red gate into the lower toggle is already available in the 0-th column, while half the Yellow gate tube of that column is available as a Yellow gate for the new column.

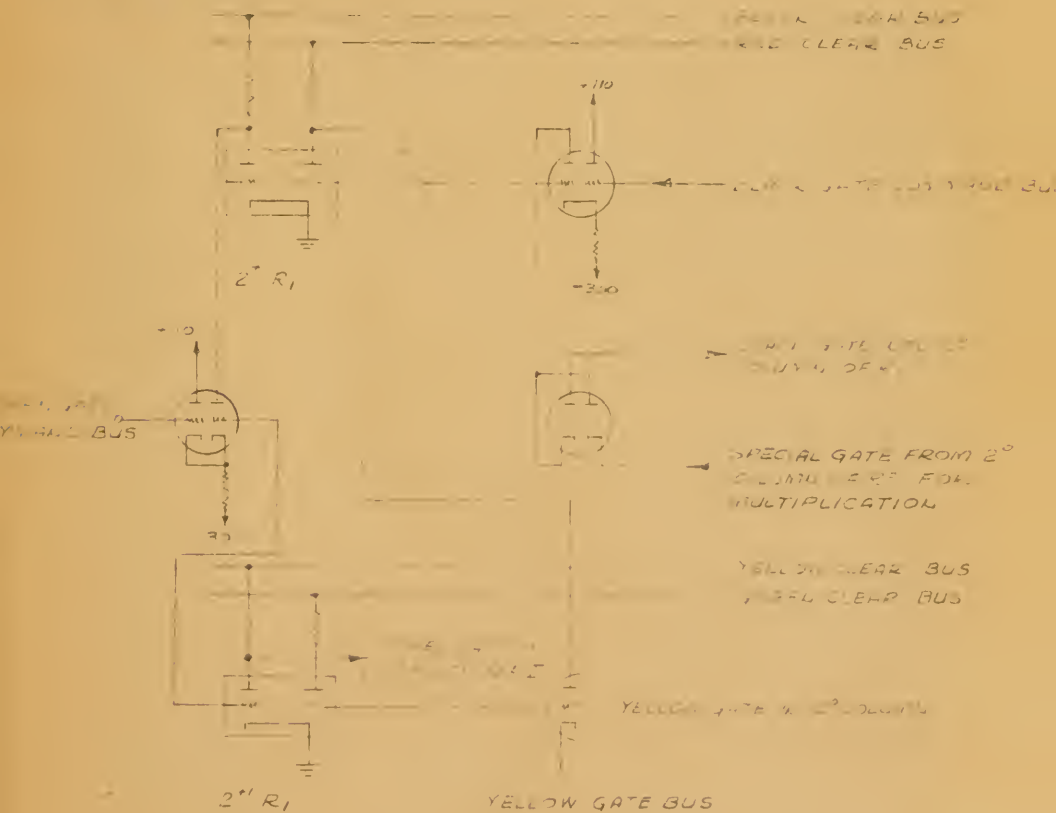


Fig. 10.

A Black gate is provided for right shifts. It will be noticed that a Green gate is also provided, which, unlike the other Green gates in RI, transmits from the lower to the upper toggle of the new column. The diode arrangement also permits the transmission of information from the 0-th column of R^3 to the corresponding column of R_1 .

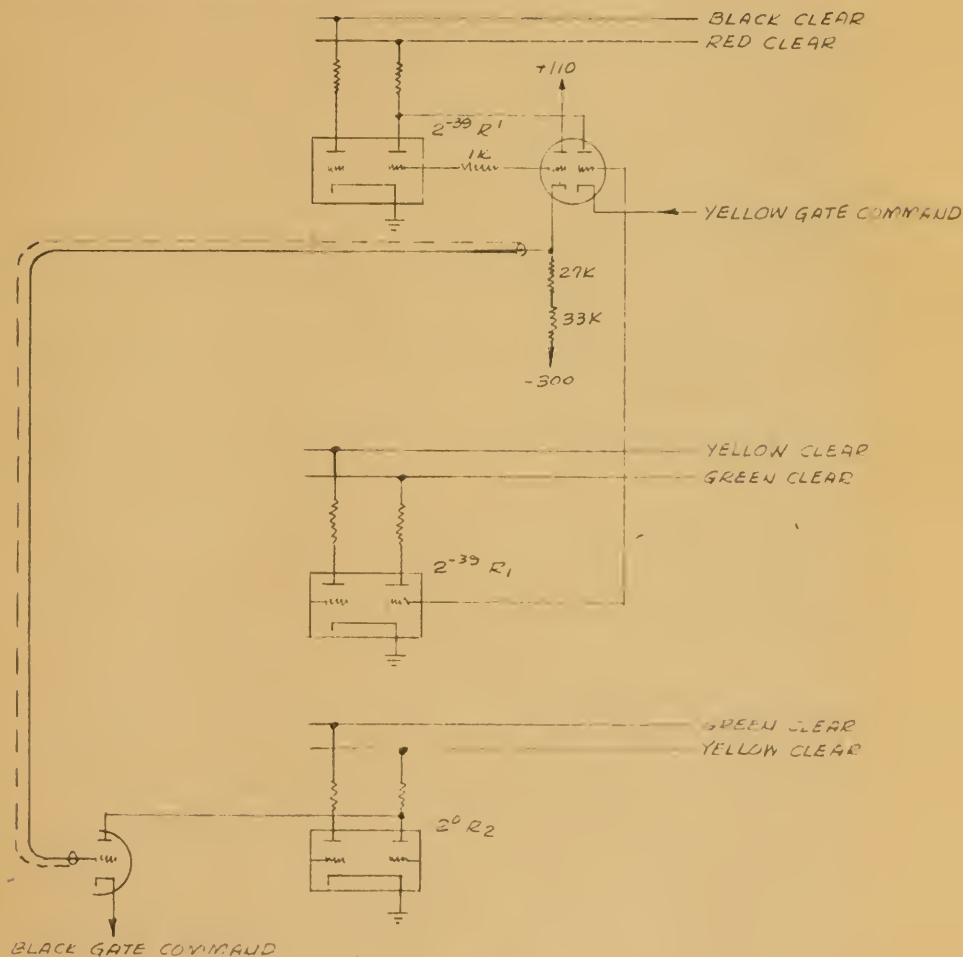


Fig. 11.

The transmission of the overflow from the 39th column of RI to the 0-th column of RII is simply taken care of by two additional tubes: One of these is a 2C51 mounted just beyond the extreme right end of RI: one section of this is used as a Yellow gate for the 39th column of RI, while the other is a cathode follower driven from one grid of the upper toggle of the 39th column, and driving a coaxial cable which runs back to the extreme left end of RII, as shown in Fig. 11. If right shifts are ordered in both RI and RII, the toggle in the 0-th position in R will have been cleared to 0 before the Black gate is opened. The opening of the Black gate will put into the 0-th stage of R_2 a 1 if a 1 was held in the upper toggle of the 39th stage of RI, otherwise there will be no change. Thus in any case the contents of $2^{-39}R^1$ is shifted into the 0-th stage of R_2 . See Fig. 11.

The performance of the clearing operations in RI and RII is the function of the Clear Selector and Clear Driver Chassis: for each Register there is one Clear Selector Chassis and four Clear Driver Chassis, one for each of the types of clear operation used.

As the term indicates the Clear Selector Chassis determines which clear is to be performed. Fig. 12 shows half the tubes in the RI Clear Selector -- those which determine whether a Red or a Black clear is to be performed. This circuit consists of a 6J6 twin triode and two 6AL5 diodes. Note that the cathode of the 6J6 is either at +10v or at -10v, determined by the Logical Control Organ of the machine, while the grids are at either 0 or -40v. Thus as long as the cathode of the 6J6 is maintained at +10v, both sections of the tube will remain cut off, while the plate voltage will be maintained at +150v by the 6AL5 diode shown in the drawing just above the 6J6. This is in turn applied to the grid of the 6J6 Clear Driver cathode follower, whose

BLACK CLEAR (R' TO O'S)
(IDENTICAL CIRCUIT)
(S) 5687's & (1) 606

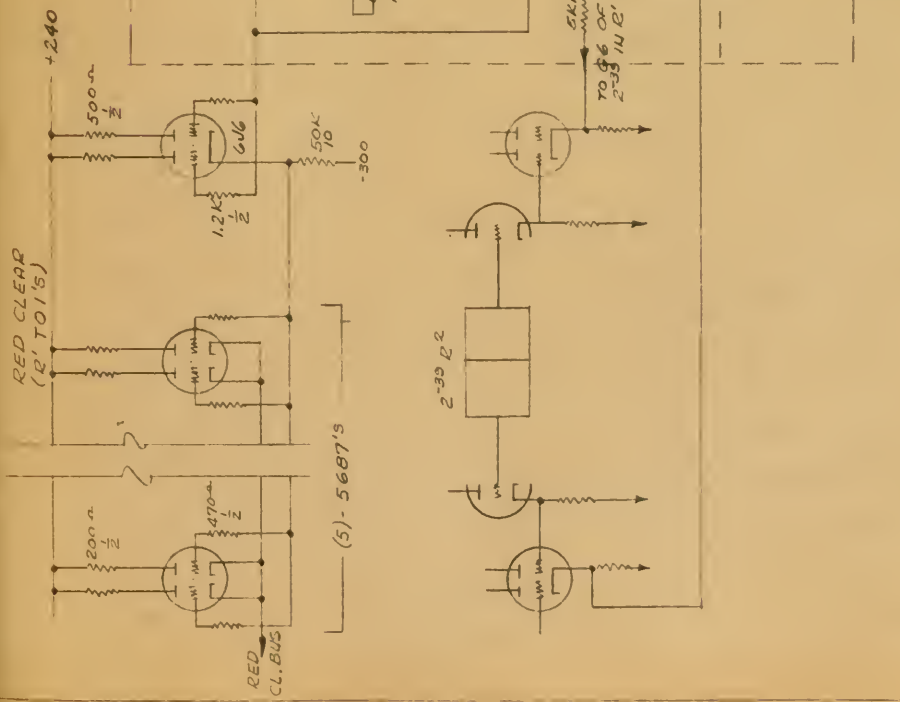
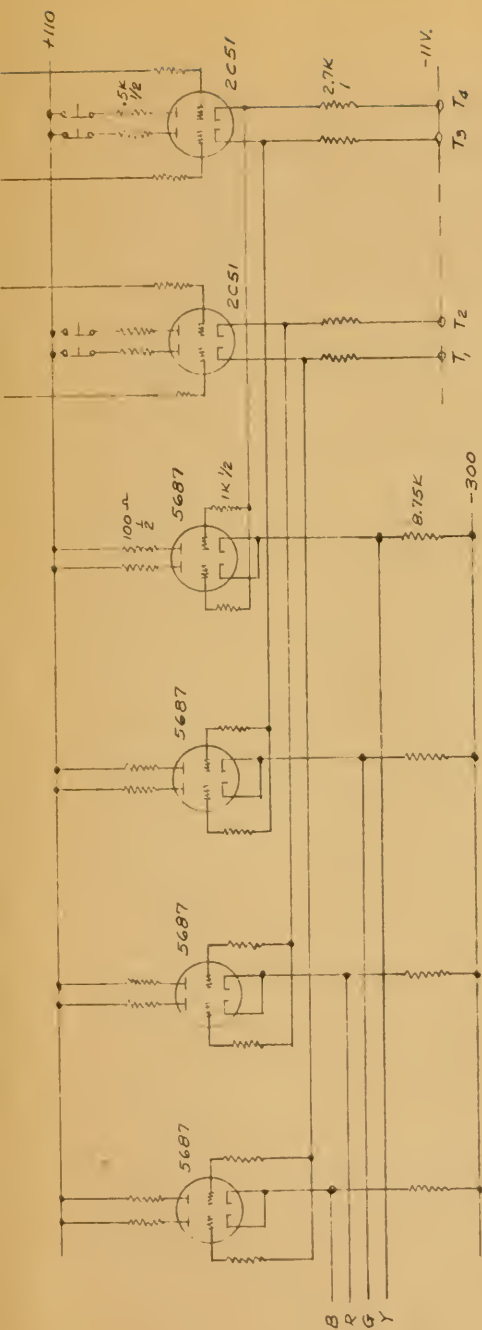


Figure 12

cathode in turn drives the grids of five 5687 tubes connected in parallel. The cathodes of these tubes are connected to the Red clear bus, which supplies plate voltage to the toggles of R^1 . These tubes, therefore, again merely form a cathode follower of large current handling capacity. We, therefore, see that as long as the 6J6 in the Clear Selector remains in a cut off condition, the Red clear bus will be maintained at +150v.

Now let one of the grids of the 6J6 Clear Selector rise to 0v: to be explicit, the left-hand one, the right-hand one remaining at -40v. Nothing happens as long as the cathode is maintained at +10v, because the tube will remain cut off. Now let the cathode be switched to -10v. The left-hand side of the 6J6 will conduct, and the cathode will follow the left-hand grid to 0v, thus holding the right half in a cut off condition. Hence the right-hand plate of the 6J6 will remain at +150v, and the Black clear bus will be maintained at this voltage. The plate of the left half of the tube will, however, because of the plate current flowing through the 39,000 ohm resistor, fall to about 50v, which will cause the Red clear bus voltage to swing down to this value also. Reversing the voltages at the two grids of the 6J6 in the Clear Selector will clearly cause the Red clear bus to remain at +150v, while the Black clear bus will swing down to +50v. Another set of two 6AL5's and a 6J6 in the Clear Selector permit the choice of Green or Yellow clear operations. The toggles to be inspected in this case will be determined later; this is also the situation with regard to the Clear Selector for RII.

The circuits supplying the inputs to the grids of the 6J6 in the Clear Selector permit the inspection of the contents of two toggles -- one is that in the 2^{-39} position in R_2 , the other is not yet determined. How the contents of these toggles are used to select the appropriate clearing operations



In RII Gate Chassis, T_1 , T_2 , T_3 are connected as shown, but T_4 is driven by the

Yes-No signal from the Control: +10 = No, -10 = Yes.

Each gate bus drives 16 grids, 4 in each chassis.

Fig. 13.

will be shown in our description of the Logical Control of the machine.

In Fig. 13 is shown the schematic of the RI Gate Drivers and Gate Driver-Drivers. Each gate command is supplied by two cathode followers in cascade: the first, which we shall call the Gate Driver-Driver, is one section of a 2C51, while the driver itself consists of the two sections of a 5687 in parallel.

The Complement Gates permit either the number in R_3 or its one's complement to be read into the Adder. These are shown in the top row of DWG No. 1323. There is one 6J6 Complement Gate for each stage; every five of these have their plates driven by two 5687's connected as shown. Suppose the number in the toggle is a 0, the left grid being high, while the right is low. The left and right grids of the 6J6 gate tube will be then at about 0v and -40v, respectively. If the left grid of the 5687 is at +90v and the right one at -30v, then certainly the cathode voltage of the 6J6 will be 0v, signifying 0 to the Adder. If these grid voltages are interchanged, the 6J6 cathode voltage falls to a level of about -40v, signifying a 1 to the Adder. Thus, this arrangement permits us to transmit to the Adder either the contents of R_3 or its one's complement.

ADDER AND THE DIGIT RESOLVER

The performance of the operation of addition is the function of the Adder and the Digit Resolver. Two numbers to be added are presented each as a set of forty voltages, 0v representing 0 and -40v representing 1, as inputs to the forty stages of the Adder. The Adder contains an analogue feature, in that voltages are added, in part, by the addition of currents in

a resistor. The output of each stage of the Adder is one of four possible voltages, corresponding to the four possible cases, in which the result of an addition is a 0 or a 1 with or without carry into the next column. The Digit Resolver is designed to discriminate between these four possible outputs, and to give as its output one of two different voltages, which are interpreted as 0 or a 1.

Each stage of the Adder has three inputs -- two representing digits of the number to be added, and one the carry from the stage which adds digits of the next lower column of the binary numbers to be added, and two outputs, one representing a carry to the next higher stage, the other being the one which the Digit Resolver must interpret as the appropriate digit of the sum.

The physical layout of the Adder closely resembles that of the Registers. The Adder thus consists of four chassis, each containing ten columns of four tubes each, one for each digit column of a forty binary digit number. The inputs are taken from R_1 (Resident digit) and from R^3 (Incident digit). In DWG No. 1334 will be found a schematic of one of the chassis, the other four being identical: only the first two and the last two columns of the chassis are shown in the drawing, as the rest are merely repetitions. It should be noticed that the left column of the extreme left chassis operates upon the lowest order digits of the numbers to be added -- that is, upon the digits in the column 2^{-39} -- and the order increases as we go to the right.

Consider the first two columns, which will be numbered n and $n+1$, in the Adder: reference should be made to the schematic, DWG No. 1334. For simplicity in reference, the tubes of column n will be labelled $T_{1,n}$, $T_{2,n}$, $T_{3,n}$, $T_{4,n}$, while those of column $n+1$ will be $T_{1,n+1}$, etc. The inputs are:

from R_1 (Resident digit) to grids 6 of $T_{2,n}$ and $T_{2,n+1}$; from R^3 (Incident digit) to cathodes 8 and 2 of $T_{3,n+1}$; carry from column $n-1$ to grid 6 of $T_{4,n}$, and carry from the n -th column to grid 6 of $T_{4,n+1}$. Thus tubes $T_{1,n}$, $T_{2,n}$, $T_{4,n}$, and the left halves of $T_{3,n}$ and $T_{3,n+1}$ perform the addition in the n -th column while $T_{1,n+1}$, $T_{2,n+1}$, $T_{4,n+1}$ and the right halves of $T_{3,n}$ and $T_{3,n+1}$ perform that in the $n+1$ -th column. The following names for the tubes are descriptive of their functions: $T_{1,n}$ - "carry gate"; $T_{2,n}$ - "resident digit gate"; $T_{3,n}$ - "summing resistor driver"; $T_{3,n+1}$ - "incident digit gate"; $T_{4,n}$ - "summing resistor driver driver".

In the interests of clarity we will consider the possible cases, and observe the action of the circuit in each case. These cases are:

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Resident digit	0	0	0	0	1	1	1	1
Incident digit	0	0	1	1	0	0	1	1
Carry from next lower stage	0	1	0	1	0	1	0	1
Sum	0	1	1	0	1	0	0	1
Carry to next stage	0	0	0	1	0	1	1	1

Thus there are seven possible inputs, but only four possible outputs.

First consider case (1). Here the Incident and Resident digits (i.e., inputs to cathode 8 of $T_{3,n+1}$ and to grid 6 of $T_{2,n}$) are both 0v, and we can consider the carry digit connection as cut -- it will appear as we go on that this is correct, as no current flows through this wire except when there is a carry from the preceding column. Under these conditions the cathode of the Summing Resistor Driver Driver is slightly above +21lv. Hence the cathode of the left half of the Summing Resistor Driver is slightly higher than this,

so that the voltage at the grid of the Summing Resistor Driver, which is directly connected to this cathode is 215v. Furthermore, the left half of the Resident digit gate will be in a conducting stage, so that by cathode follower action the cathode will be nominally at the grid potential, or 0v, and the right half will, therefore, be cut off, since its grid is maintained at -9v. The left half of the Incident digit gate will also be cut off, since its grid is at -9v and its cathode at 0v. Thus no current will be drawn through the Summing Resistor (connected between the Summing Point and the cathode of the Summing Resistor Driver) by either of these gate tubes. The cathode of the carry gate will follow the more positive grid and, therefore, rise to 215v, so that the left half of this tube will be in a conducting state and the right half cut off. Hence the voltage at point S will be +215v, as no current flows through the Summing Resistor, and no current flows through the carry connection to the next stage, showing the correctness of our original assumption.

In cases (3) and (5), we still consider the carry into the n-th stage as disconnected. In case (2) the Incident digit input drops to -40v, which permits conduction by the Incident digit gate; the cathode of this tube will, of course, only drop to slightly above -9v. Thus the plate current drawn will be very nearly $165/32 \text{ ma} = 5.15 \text{ ma}$, which flowing through the Summing Resistor will cause a drop in voltage across this resistor of very slightly more than 54 volts. As the right half of the Resident digit gate is cut off, the output voltage to the Digit Resolver is +161v. In case (4), the Incident digit gate is cut off, but the right half of the Resident digit gate is now conducting; again the plate current of $165/32 \text{ ma}$ causes a drop of 54 volts in the Summing Resistor, so that the output voltage is again +161v. In both cases (2) and (4) we see that the left grid (6) of the carry gate tube ($T_{1,n}$)

being at +157v, the cathode of this tube will be slightly more positive, and the right half will remain cut off. Thus we have shown that the condition of "no carry" implies that no current will flow through the "carry" connection to the next stage, which is the assumption that has been used all along in the analysis.

It has now been shown that if the Incident or Resident digit is a 1, the circuit will cause the corresponding gate to draw 5.15 ma through the Summing Resistor. Hence in case (7) a total of 10.3 ma will flow through this resistor, and the output voltage to the Digit Resolver will drop to +107v. Thus grid 6 of the carry gate will drop below grid 5, which means that the right half section of this tube will now conduct, the cathode will rise to slightly more than +136v, the left section will be cut off and the right section will draw very nearly $436/90 = 4.85$ ma. This state, which indicates a carry into the next stage, causes the grid of $T_{4,n+1}$ to drop to approximately +124v.

Now consider case (2). Here grid 6 of the Summing Resistor Driver ($T_{4,n}$) is held at +124v; the diode-connected right half has its plate maintained at +161v; hence this section will conduct, and the cathode voltage will rise to slightly below +161v. This will, therefore, fix the voltage of the cathode of the Summing Resistor Driver. (At slightly below this level; there is, however, a slight rise in voltage from grid to cathode of the Summing Resistor Driver, so that the voltage at this point turns out to be very close to +161v.) Since no current is drawn through the Summing Resistor, the Summing Point voltage (V_s) is +161v. This is also sufficient to raise the cathode of the carry gate to +161v, and to keep the right half of this tube in a non-conducting state.

The remaining cases can now be discussed. In cases (4) and (6) the action of the carry from the preceding stage is to lower the cathode voltage of the Summing Resistor Driver to +161v, while the effect of the Incident or Resident one, as the case may be, is to cause a voltage drop of 54 volts across the Summing Resistor. Thus the output $V_g = 161 - 54 = 107v$. Hence the right half of the carry gate will conduct, and so a carry will be indicated to the next stage.

Only case (8) now remains. Here the Summing Resistor Driver cathode voltage is +161v because of the carry from the preceding stage, while the current drawn by the Incident and Resident gates through the Summing Resistor causes a drop of 108 volts across the Summing Resistor, so that $V_g = 161 - 108 = 53v$.

To sum up, there are four possible results to the addition in the n-th column: the table gives these, and the corresponding V_g .

Sum	Carry	V_g	V_g (average measured values)
0	0	215 volts	215.3 volts
1	0	161 volts	161.0 volts
0	1	108 volts	108.4 volts
1	1	53 volts	55.9 volts

It is now necessary to provide a means by which these output voltages are interpreted; the result of this interpretation must be that $V_g = 215$ or 108v ultimately set a toggle in the 0 position, while $V_g = 161$ or 53v set it in the 1 position. We need not worry about carries, as we have seen that these are all automatically taken care of in the Adder. The task of interpretation is assigned to the Digit Resolver, which will now be described.

Physically the Digit Resolver consists of four chassis similar to those used in the Adder and located directly above them. Each column of tubes in the Digit Resolver is assigned to interpret the output of one column of tubes in the Adder. Thus the columns of tubes in the Digit Resolver are independent of each other, accepting a single input (V_g) from the Adder, and giving a single output to R^1 . A schematic of a single column of tubes in the Digit Resolver is shown in DWG No. C-3-1107. For convenience these will be labelled $T_1, \dots, T_4$ in order from the top downward.

We shall consider in order the four possible cases. First let $V_g = +215v$, which being applied to grid 5 of T_3 causes the cathode to rise slightly higher, thus cutting off the right section of T_3 and the left section of T_4 . Furthermore, the cathode of T_1 is raised above $+215v$, cutting off the right section of that tube. Now consider T_2 . If the plate and grid of the left section were disconnected, grid 3 would assume a potential of approximately $+163v$. Actually, of course, grid current flows, so that the grid is only slightly positive with respect to the cathode. Plate current amounts to about 4 ma which is more than sufficient to hold grid 7 below cut off. Thus the output voltage is determined by the voltage divider consisting of the 43.16K, 128.7K, and 188.1K resistors to be $+32v$.

In the second case, $V_g = +161v$. This is still sufficient to hold the right section of T_3 in a non-conducting state, but not enough to do the same for the left section of T_4 , which accordingly conducts, while the right section is cut off. Again the voltage of grid 6 of T_1 ($+161v$) is more than enough to assure that the right section is cut off. As the left section of T_4 draws approximately $488/139 \div 3.5$ ma, the voltage of grid 3 of T_2 falls to about $+86v$, and the left section of the tube is, therefore, cut

off. The voltage of grid 7 of T_2 would now become +116v but for the flow of grid current, which holds it down to but little above +82v (the voltage of cathode 8), and the output voltage falls to about -58v.

In the third case, $V_g = +108v$. Again the left sections of T_3 and T_4 conduct, while the right sections are cut off. Hence as in the second case the grid voltage of the left section of T_2 is well below cut off. However, the voltage of grid 6 of T_1 is now but slightly above +108v, while that of grid 5 is +136v: the cathode follows the more positive grid, the left section is cut off, and the right section draws a plate current of very nearly 4 ma. As in the first case, this holds the voltage of grid 7 of T_2 safely below cut off, and again the output voltage is approximately +32v.

Finally, in the fourth case $V_g = +56v$ (to use the average measured value). Hence in T_3 the right section conducts, while the left section is cut off. The right section of T_4 and the left section of T_1 are cut off, while the other sections conduct, and consequently, as in the third case, both sections of T_2 are cut off. Now, however, the plate current drawn by the right section of T_2 through the 43.16K resistor causes the output voltage to fall to about -48v.

To sum up, the Digit Resolver output is +32v in both cases where the sum digit is 0, while it is far negative in both cases where the sum digit is 1. These voltages are actually considerably larger than needed to operate the RI Green gates. Hence a diode "bumper" is provided at the output of each stage: the circuit is shown in DWG No. 1288. In stages 2^0 to 2^{-35} , the diode cathodes are returned to ground, while in the remaining stages they are returned to ground except during the read-in of information from the Input. Thus during ordinary operation, the Digit Resolver output never swings above 0v, while the negative excursion is not limited.

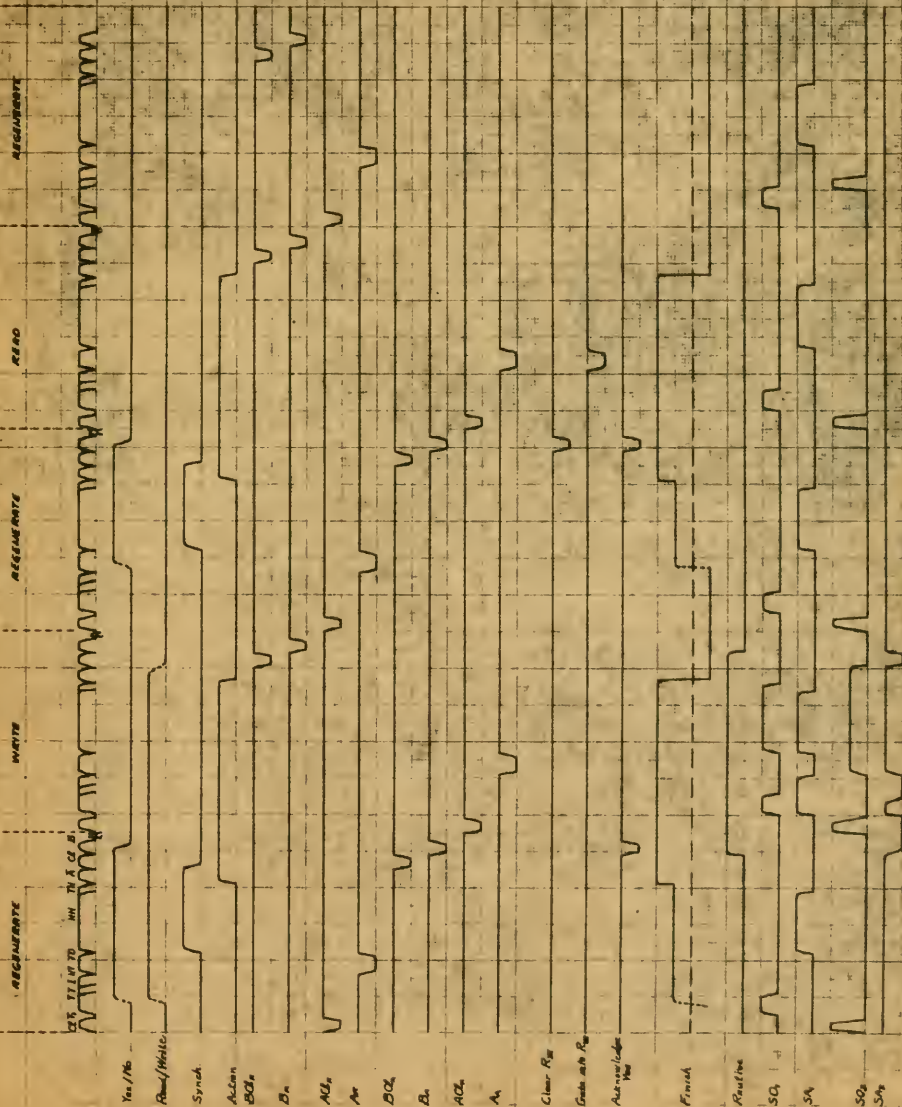
our sequence of events, after 8) all the toggles have been set to their off condition and will remain so at least until after 10), since if an Action cycle is to be called for, the Synch toggle cannot be turned on until TD_t .

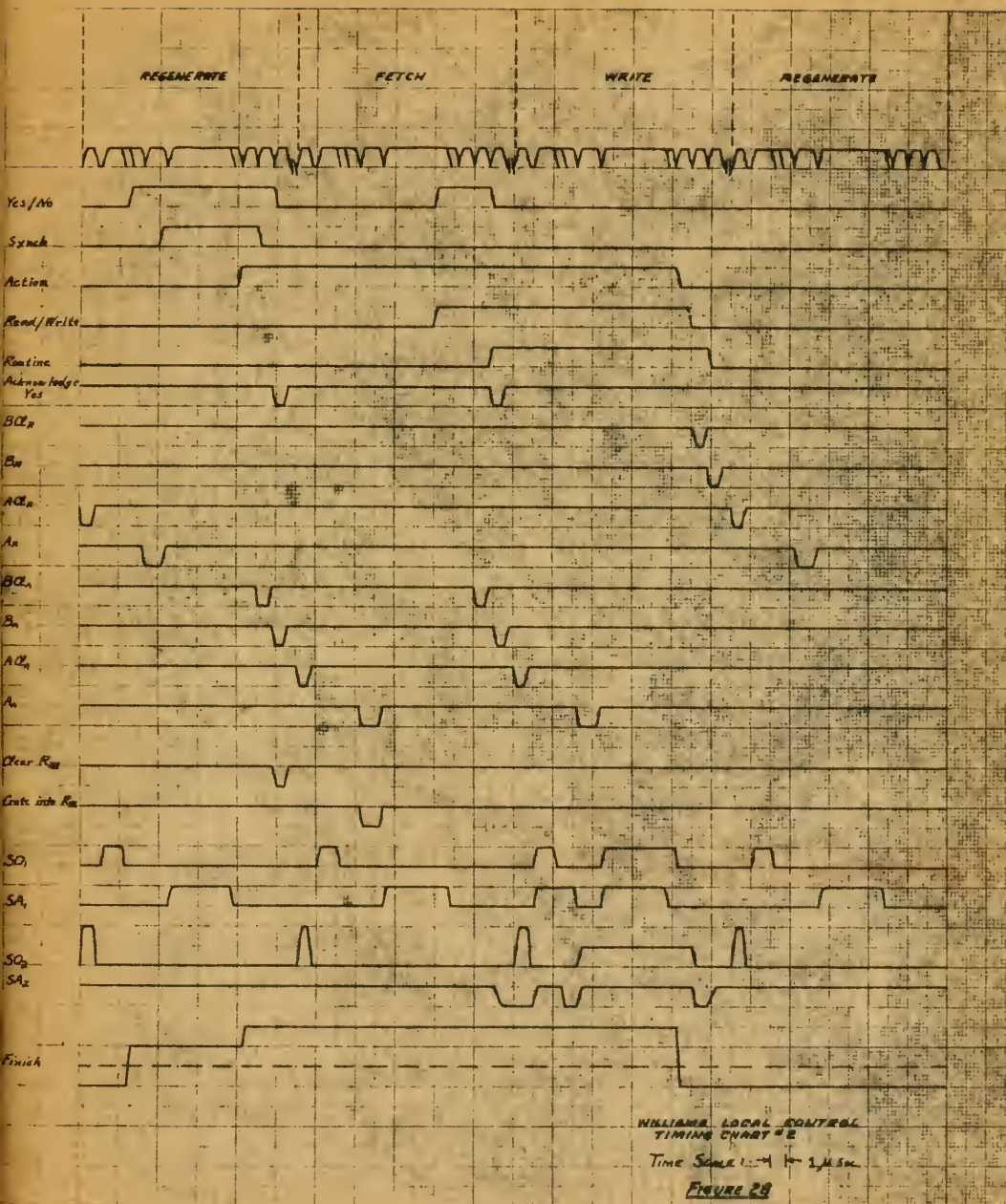
Complete diagrams are given in Figures 27 and 28 illustrating the operations that have just been described. Fig. 27 shows the time variation of the voltage inputs and outputs for the following sequence of cycles: Regenerate, Write, Regenerate, Read, Regenerate. Fig. 28 does the same for the sequence: Regenerate, Fetch, Write, Regenerate.

IV. THE MAIN CONTROL

We have described the Williams Memory in some detail, and have shown how the Local Control regulates its operation subject to information received from the Main Control. This information must be of two kinds. First, it is necessary to specify whether an Action or a Regenerate cycle is to be performed, and whether information is to be inserted in or extracted from the Memory. Second, it is necessary to supply to the deflection circuits the address in the Memory of the bits of information which are to be written, read or regenerated. The equipment needed to decode that part of each order which specifies the Yes/No and Read/Write alternatives was not yet completed while this report was being written. The Dispatch Counter, which supplies address information to the deflection circuits, was completed and installed, and will be described below.

Besides supplying orders and address information to the Memory Organ, the Main Control also must regulate the actions of the Input-Output Organ and of the Arithmetic Organ. Most of the equipment for the latter function has been built and installed, and will be described under the headings "Gate-Clear Sequencing Chain", "Shift Counter", and "Recognition Circuits". Obviously





our account of these circuits will be rather fragmentary, and it will be necessary to show in a later report, once the remaining portions of the Control have been completed, how the separate parts are connected into an integrated unit.

THE SHIFT COUNTER

Two types of counter are used in the present machine. These are the familiar "scaling" type, of which the Shift Counter is an example, and the "adder" type, which operates as an Adder to which one input is the last count and the other is a 1 to the lowest order stage. The Scaling type of counter has the advantage that the more significant stages can be comparatively slow, and that all the carries resulting from one count need not be completed before the next count arrives at the input, whereas in the Adder type, all carries resulting from one count must be completed before the next count is added. On the other hand, the Adder type offers the advantage of requiring fewer tubes. Hence, the character of the particular application indicates the appropriate type of counter. In the Dispatch Counter the complete new address must be obtained before the counter is stepped again, so that the Adder type is obviously appropriate, while in the Shift Counter this is not true, so that the Scaling type is used.

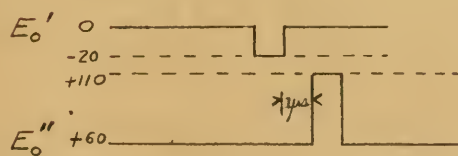
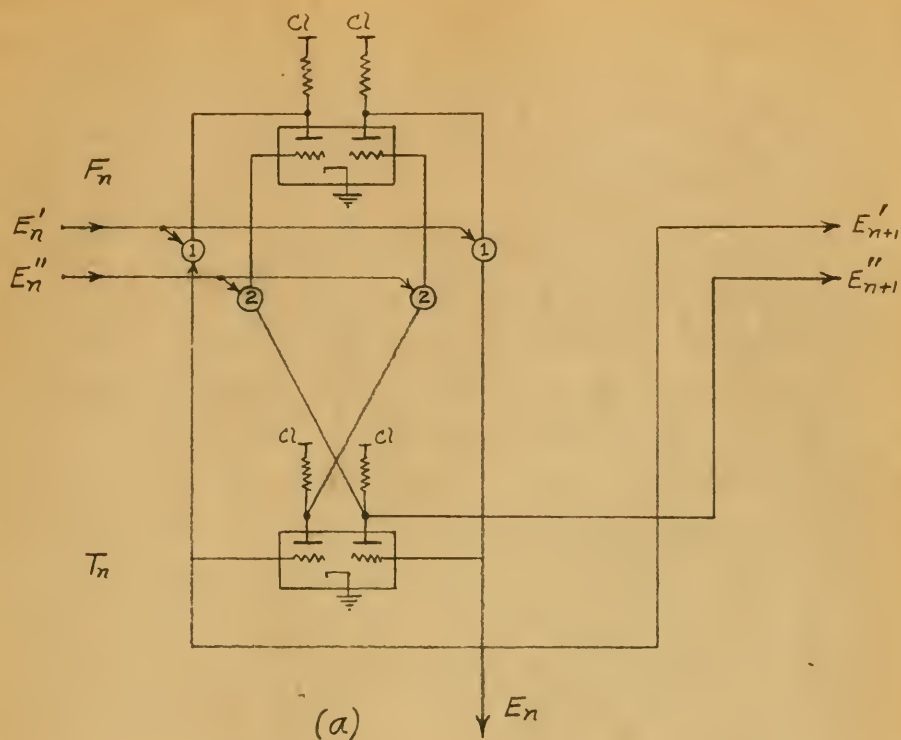
The Shift Counter is part of the equipment used to control the carrying out of the arithmetical processes. Thus multiplication is performed by successive additions and shifts. Once the process is initiated, the Shift Counter counts the successive shifts. The number of these which must be performed is inserted in the Recognition circuit, where it is continuously compared with the contents of the Shift Counter; when coincidence is indicated,

a signal is emitted which terminates the process. A similar situation occurs in the carrying out of division.

Physically the Shift Counter consists of two rows of toggles with gates permitting information to be transferred up and down vertically and up diagonally. One row of toggles holds the actual count, the other does not; these are referred to as the True and False rows, and the numbers they hold as the True and False counts, respectively. It will appear that the False count does not proceed monotonically. There are six stages in the counter, those for 2^0 , 2^1 , ..., 2^5 . Since the counter must only count to forty (the number of additions in the multiplication process) the 2^5 stage can be rather simpler in structure than the lower order stages; it is feasible to have but a single toggle in this sixth stage.

Before examining the schematic of the counter, let us consider the block diagram of a single stage, given in Fig. 29(a). This represents any of the stages 2^0 to 2^4 . It has already been remarked that the sixth (2^5) stage is of simpler structure; it will be considered later. Here input E_n' enables gates from the True to the False toggle, which are so arranged that the enabling of the gates causes F to assume the same condition as T. On the other hand, input E_n'' enables gates leading from F to T in such a way that the enabling of the gates causes T to assume the condition opposite to that of F. At the input to the 2^0 stage, the levels for E_0' are 0v to disable, -20v to enable, while those for E_0'' are +60v to disable, +110v to enable; in the following stages the levels of E_n' become 0v and -35v, and those of E_n'' remain substantially unchanged. We will have more to say on this subject when we discuss the circuitry.

The input to the lowest order (2^0) stage of the counter consists of



INPUT TO 2° STAGE

(b)

FIGURE 29

the pair of pulses shown in Fig. 29(b) for each item to be counted. The number held by the counter is presented to the recognition circuit, which is physically located on the same chassis, as a set of six voltages, one for each stage of the counter, the presence of a 0 being signified by 0v, that of a one by -40v.

Each stage of the counter, therefore, has two inputs and three outputs; on Fig. 29(a) the inputs are designated E'_n and E''_n , the outputs E_n , E'_{n+1} , E''_{n+1} . Of these E_n signifies to the recognition circuit the contents of the stage, while E'_{n+1} , E''_{n+1} are inputs to the next stage. In the actual circuit, DWG No. 1289, let us label the tubes T_{ij} , where i signifies the row and j the column in the array. The first five columns and the lower three tubes of the sixth form the actual counter. The upper four tubes in the sixth column are clear drivers used to clear the two rows of toggles. Consider the first column. T_{11} is the F toggle of the block diagram, T_{21} is the up gate, T_{41} the down gate, T_{31} the gate driver, while the lower three tubes, T_{51} , T_{61} , T_{71} are the T toggle, which is actually what we have before termed a Supertoggle. It is used in this instance because of the speed with which it can be caused to switch from one state to the other.

The counter is prepared for operation by clearing all the T toggles to 0 and the F toggles to 01111 (2^0 holding a 0, the others 1's). Consider the first column. The first pulse of the input pair clearly does nothing, for in this case both T and F already hold 0's. On the other hand, the down gates transmit a 1 to T if F holds a 0. Hence after the reception of the pair of pulses T holds a 1.

Now let us consider the influence of the n-th column of the counter

upon the $(n+1)$ -th. According to the block diagram of one stage, the gating voltages (1) and (2) to the next stage are the left grid and right plate voltages of the T_n toggle tube. Hence while this toggle holds a 0, (1) and (2) are both at their high level with the result that in the next column the up gates are disabled and the down gates enabled. On the other hand, when T_n holds a 1, the gating voltages are both held at their lower values, with the result that the up and down gates of the next stage are, respectively, enabled and disabled.

Thus if T_n holds a 0, T_{n+1} must hold a 1 or a 0 if F_{n+1} holds a 0 or a 1, while if T_n holds a 1, the contents of F_{n+1} must coincide with that of T_{n+1} . These facts mean that whenever T_n holds a 0 its next change of condition to 1 cannot change T_{n+1} , but does cause F_{n+1} to agree with T_{n+1} , while whenever T_n holds a 1, its next change of condition to 0 cannot affect F_{n+1} , but must cause a change of condition in T_{n+1} . Thus two successive changes of condition in T_n cause a single change of condition in T_{n+1} , so that the performance of the circuit as a binary counter of the scaling type is guaranteed.

We will now consider the actual circuits involved, and in particular will show how the simplified sixth stage of the counter works, and how its simplified form restricts the number of items that can be counted without impairing the usefulness of the counter in its particular application.

The toggles of the False rank and that in the sixth stage of the True rank are of the type encountered throughout the computer, while the first five toggles of the true rank are the so-called Supertoggles. The gating arrangement from T to F is quite straightforward; we describe it for the first column, the remarks obviously also applying to all others but the sixth. Each gate consists of one section of T_{21} and the left section of T_{31} . From

the way in which the grids of the toggle tube T_{61} are returned to $-300v$, it is readily seen that when the voltage of a grid of the toggle tube is $0v$, that of the corresponding grid of the gate tube T_{21} is approximately $-10v$, while when the toggle grid voltage is $-40v$, that of the gate grid is approximately $-48v$. Hence, if the grid of the left section of T_{31} is held at $0v$, it is clear that the cathode of T_{21} is held so far above that of either grid that conduction is impossible, while if the grid voltage of the left section of T_{31} is dropped to either $-20v$ or $-40v$, that section of T_{21} will conduct, the grid voltage of which is $0v$, while the other section will remain cut off.

Now consider the "down" gates, each composed of the right section of T_{31} and one section of T_{41} . The grid voltages of T_{41} are either $0v$ or $-40v$, while its cathode is returned to ground, and both plates are connected to the right cathode of T_{31} through $5.6K$ resistors. While the voltage of the right grid of T_{31} is held at $60v$, that of the plate of the conducting half of T_{41} is substantially below $60v$ (apart from cathode rise in T_{31}). The plates of T_{41} are connected to the left grid of T_{51} and the right grid of T_{71} , the other grids of these tubes being connected to the plates of the Supertoggle tube, T_{61} . Hence, while the gating voltage E_0 is held at $60v$, neither plate of T_{41} is effective in determining the cathode voltage of T_{51} or T_{71} , and the Supertoggle can remain in either condition.

Now suppose the voltage (2) to be raised to $+110v$, bringing with it the voltage of the plate of the cut-off section of T_{41} . Let us assume that F_1 holds a 0, so that this is the right section of T_{41} : the plate voltage of the left section also rises to about $60v$. Suppose now that T_1 holds a 1, so that the grid voltage of the right section of T_{51} is about $+60v$, and that of the left section of T_{71} is $+110v$. The grid voltages of the left section

of T_{51} is about +60v, and that of the left section of T_{71} is +110v. The grid voltages of the left section of T_{51} and the right section of T_{71} are also +60v and +110v, respectively, so clearly nothing happens. On the other hand, if T_1 holds a 0, the grid voltages of the right section of T_{51} and the left section of T_{71} are +110v and +60v, respectively, and, therefore, the cathode voltage of T_{71} is raised to +110v, the voltage of the right grid. Clearly this flips the supertoggle to the 1 position. Similarly, if F_1 holds 1 and T_1 0, the raising of the gating voltage E''_0 from +60v to +110v does not affect T_1 , but if T_1 holds 1, it causes it to flip to 0.

The discussion just given applies to the columns $2^0, \dots, 2^4$. The 2^5 column is simpler in structure, consisting of three 2C51 tubes: T_{56}, T_{66}, T_{76} , of which T_{66} is the toggle tube, the other two performing gating functions. Inputs are: from the left grid of T_{65} to the left grid of T_{76} , and from the right grid of T_{65} to the left grid of T_{56} .

Suppose 0's exist in the 2^4 and 2^5 stages. Then the left section of T_{56} is cut off, and the right conducting, the cathode voltage being -40v. The left section of T_{76} is conducting, the cathode being 0v. Now consider the right section of T_{76} , which is connected as a diode. The plate is connected through a 4.7K resistor to the cathode of the left section, and the cathode is connected directly to the cathode of the right section of T_{56} . Hence, the right section of T_{76} conducts, and its plate voltage, which is the output from this stage to the recognition circuit, is not much above -40v.

Now let the 2^4 stage toggle flip to 1. The cathode of the left section of T_{76} now falls to -40v, and as the cathode of the right section cannot fall below this voltage, the output remains at -40v. The left section of T_{56} conducts, and the usual gating action takes place, the toggle being flipped to the 1

condition. Thus flipping the toggle in the 2^4 stage to 1 automatically flips the toggle in the 2^5 stage, but does not effect the output voltage, which remains -40v, signifying a 1.

This condition persists as long as the 2^4 stage holds a 1, hence, up to and including the count of 31. On the count of 32, stages 2^0 , ..., 2^4 must all return to 0. The left section of T_{56} is now cut off leaving the toggle in the 1 condition, and the cathode of the right section, and hence the cathode of the right section of T_{76} , rises to 0v. At the same time the cathode of the left section of T_{76} rises to 0v, and we are assured that the right section of T_{76} will not conduct, and its plate voltage, therefore, becomes 0v, signifying a 1 to the recognition circuit.

The lower order stages now proceed with the count, the 2^4 stage remaining in the 0 condition until the count of 48, when it flips to 1. We have already seen, however, that if both the 2^4 and 2^5 stage toggles are in the 1 condition, the 2^5 stage output voltage drops to -40v, signifying a 0 to the recognition circuit. The shift counter can only count to 47, returning to 32 on the count of 48. However, since the Shift Counter is only required to count to 40, this is no limitation on its usefulness. The recognition of this fact permits the simplification of the 2^5 column, with attendant saving in tubes. The space thus saved is occupied by the two clear drivers T_{16} , T_{36} , and the cathode followers T_{26} , T_{46} : T_{26} permits all the plate current for the F toggles to be drawn from the +240v bus, thus holding this current constant and preventing any fluctuation of plate voltage due to current fluctuation in the power supply and bus impedance, while T_{46} accomplishes the same for the T toggles and the +220v bus.

Fig. 30 shows the contents of the F and T toggles as the counter proceeds from its quiescent condition to the count of forty. In the T_5 column

T_0	T_1	T_2	T_3	T_4	T_5	F_0	F_1	F_2	F_3	F_4
0	0	0	0	0	0	0	1	1	1	1
1	1	0	0	0	0	0	0	1	1	1
2	0	1	0	0	0	1	0	0	1	1
3	1	1	0	0	0	0	1	0	1	1
4	0	0	1	0	0	1	1	0	0	1
5	1	0	1	0	0	0	0	0	0	1
6	0	1	1	0	0	1	0	1	0	1
7	1	1	1	0	0	0	1	1	0	1
8	0	0	0	1	0	0	0	0	1	0
9	1	0	0	1	0	0	0	0	1	0
10	0	1	0	1	0	0	1	0	0	0
11	1	1	0	1	0	0	0	1	0	0
12	0	0	1	1	0	0	1	1	0	0
13	1	0	1	1	0	0	0	0	1	0
14	0	1	1	1	0	0	1	0	1	0
15	1	1	1	1	0	0	0	1	1	0
16	0	0	0	0	1	[1]	1	1	1	0
17	1	0	0	0	1	[1]	0	0	1	0
18	0	1	0	0	1	[1]	1	0	0	1
19	1	1	0	0	1	[1]	0	1	0	1
20	0	0	1	0	1	[1]	1	1	0	0
21	1	0	1	0	1	[1]	0	0	0	0
22	0	1	1	0	1	[1]	1	0	1	0
23	1	1	1	0	1	[1]	0	1	1	0
24	0	0	0	1	1	[1]	1	1	1	0
25	1	0	0	1	1	[1]	0	0	1	0
26	0	1	0	1	1	[1]	1	0	0	1
27	1	1	0	1	1	[1]	0	1	0	1
28	0	0	1	1	1	[1]	1	1	0	1
29	1	0	1	1	1	[1]	0	0	0	1
30	0	1	1	1	1	[1]	1	0	1	1
31	1	1	1	1	1	[1]	0	1	1	1
32	0	0	0	0	0	1	1	1	1	1
33	1	0	0	0	0	1	0	0	1	1
34	0	1	0	0	0	1	1	0	0	1
35	1	1	0	0	0	1	0	1	0	1
36	0	0	1	0	0	1	1	1	0	1
37	1	0	1	0	0	1	0	0	0	1
38	0	1	1	0	0	1	1	0	1	0
39	1	1	1	0	0	1	0	1	1	0
40	0	0	0	1	0	1	1	1	0	0

FIGURE 30

the symbol $[1]$ is used to indicate that, though the toggle is in the 1 condition, the output is so arranged as to present a 0 to the recognition circuit.

RECOGNITION CIRCUIT

The function of the Recognition Circuit is to compare the number in the True rank of toggles in the Shift Counter with some predetermined number, to determine when the two coincide, and at that instant to emit a signal which can be used to terminate the process the number of steps in which is being counted.

Since the "number to be recognized" and the number in the counter are both six place binary numbers, there are twelve inputs to the Recognition Circuit. Six of these are the output leads from the True rank of Counter toggles, and six come ultimately from the read-out gates of R_3 . There is a single output, the signal emitted when the number in the counter has increased until it coincides with the "number to be recognized."

The Shift Counter chassis contains the Recognition Circuit, and also the Address Dispatch Gates, which will be described later on. We refer to DWG No. 1289 for the schematic.

The Recognition Circuit itself consists of the the last six tubes in each of the two lowest rows in the drawing: T_{67} , ..., $T_{6,12}$, and T_{77} , ..., $T_{7,12}$. The inputs labelled (A) are clearly from the Shift Counter, while the voltages representing the "number to be recognized" are brought down on the leads labelled (B), these voltages being developed in the cathode circuits of the input cathode followers $T_{2,8}$, ..., $T_{2,12}$.

Each of the tubes $T_{2,8}$, ..., $T_{2,12}$ is a 2C51, the two sections functioning independently as cathode followers. We observe that each grid is

connected to the +110v bus through a 7.5K resistor and to the input lead from R_3 . The 110v bus is the plate supply of the RIII Green gates. Since all ten cathode follower circuits are identical, let us consider one in detail.

Suppose, for example, that the 2^0 toggle in R_3 contains a 0. Then the opening of the RIII read out (Green) gate has no effect on $T_{2,8}$, since no current is drawn from the +110v bus. The voltage of the left cathode is +110v, and, if the grid of the left section of $T_{3,8}$ were not connected to the point between the 15K and the two 22K resistors in the cathode circuit, the voltage of this point would be approximately +6v. With the grid connected as shown, a slight grid current is drawn, and the voltage of the point is reduced nearly to 0v; we consider it to be 0v as this is sufficiently accurate for our purpose.

If we consider the effect of a 5% tolerance in the values of the 15K and the two 22K resistors, it turns out that without the grid connected, the voltage at this point, assuming that the cathode of $T_{2,8}$ is +110v, can be as low as -2v and as high as +13v. Hence with the grid connected, no grid current is drawn in the first extreme case, while in the second, about twice the nominal value is drawn. Thus one would expect considerable variation in plate voltage from tube to tube for a 0 input: measured values run from 9v up to 14v with the average of 12.8v. This variability, however, is of no consequence: we will see in our discussion of the Address Dispatch Gates that the plate voltage under discussion must only fall below about 20v for satisfactory operation.

On the other hand, if the 2^0 stage of R_3 contains a 1, the opening of the RIII read out (Green) gate causes a current of 5 ma. to be drawn from the +110v bus, which reduces the voltage of the left grid of $T_{2,8}$ to +72.5v. The

voltage of the left grid of $T_{3,8}$ accordingly falls to a nominal value of -22v, cutting off the left section of the tube. The actual value is, of course, different from the nominal one for two reasons: cathode rise in $T_{2,8}$ and permitted tolerances in the 15K and the two 22K resistors. Measured values range from -17.5v to -20.3v at the ten grids of $T_{3,8}$, ..., $T_{3,12}$. Actually even -17.5v is about twice the value needed, as we shall see.

We have seen, therefore, that if a 0 is held in the 2^0 position of R_3 , the nominal voltage on the corresponding lead (B) is 0v, while if a 1 is held there, the nominal voltage becomes -22v.

Let us now turn our attention to the Recognition Circuit proper. We suppose some "number to be recognized" has been chosen, the voltages representing its digits (0v for 0, -22v for 1) being applied as inputs to the grids of the right-hand sections of $T_{7,7}$, ..., $T_{7,12}$. These inputs are established with the counter in the quiescent state, all outputs (A) representing 0's. We assume that the "number to be recognized" is not 0. Therefore, at least one input (B) is -22v, and this voltage is assumed by at least one of the cathodes of tubes $T_{7,7}$, ..., $T_{7,12}$; hence, at least one section of the 6AL5's $T_{6,8}$, $T_{6,10}$, $T_{6,12}$ is in a conducting state, with plate voltage slightly above -22v, and either the right section of the 2C51, $T_{6,7}$, or one section of the 6J6, $T_{6,11}$, is cut off, with plate voltage, therefore, at +110v. From the tube characteristics it can be seen that for both types used here, with grounded cathodes, 10K plate load resistors, and +110v plate supply voltages, cut off occurs when the grid is dropped slightly below -4v. Thus the -22v available is more than ample, and even -10v would provide an adequate margin of safety. Therefore, either the left section of $T_{6,7}$ is conducting with cathode voltage at +110v or one section of $T_{6,9}$ is conducting with cathode

voltage at +110v; in either case the voltage of the point labelled "Out" in the drawing must be +10v.

Let the counting process now begin. At the count of 1 the cathode of $T_{7,7}$ must follow the left grid to 0v. If the "number to be recognized" is 10000, then all the remaining cathode voltages are already 0v, and with the cathode of $T_{7,7}$ also assuming this value, none of the diodes conduct, all their plate voltages assuming the value 0v. Then the right half of $T_{6,7}$ and both sections of $T_{6,11}$ conduct, their plate voltages falling to about +60v in all cases, which causes the voltage "out" to fall to -27v.

Now consider an arbitrary "number to be recognized". From the previous paragraph it is clear that when the number in the counter reaches agreement with it, the voltage "out" assumes the value -27v. We must still show that the "out" voltage cannot drop to -27v until agreement is reached.

Consider two binary numbers $n > m$; if they are not of the same number of digits, we fill in the missing higher order columns of m with 0's -- this is exactly what occurs in the case under discussion. First we compare the highest order column: n certainly contains a 1 while m may contain a 1 or a 0. If m contains a 1 there, we examine the column of next lower order. Here either both numbers have a 1 or both have a 0, or if there is disagreement, n must have a 1 and m a 0 -- the contrary case is impossible, for then we have $m > n$. Hence, as we move from the highest order columns on down, the first disagreement we meet must always be between a 1 in the larger number and a 0 in the smaller. In columns of lower order than that in which the first disagreement is found, obviously anything can happen.

Physically the last paragraph means that as long as the "number to be recognized" exceeds the number in the True rank of the counter, at least one

of the tubes $T_{7,7}$, ..., $T_{7,12}$ will have its left grid held at $-40v$ and its right grid at $-22v$, thus assuring a voltage of $+10v$ at "out".

If the counter were permitted to count beyond the "number to be recognized" the voltage "out" could assume either the $-27v$ or the $+10v$ level in different cases, resulting in complete ambiguity. However, there is obviously no trouble here, as there is no point in continuing the count beyond the number to be recognized which will not exceed forty, and the "out" voltage is available to terminate the process, the number of steps in which is being counted, as soon as agreement is reached.

THE ADDRESS DISPATCH GATES

These are the tubes $T_{4,8}$, ..., $T_{4,12}$ and $T_{5,8}$, ..., $T_{5,12}$ in DWG No. 1289. It has been shown in the discussion of the Recognition Circuit that, for example, of 2^0 in R_3 is 0, the left grid of $T_{5,8}$ assumes a value between $+9v$ and $+14v$, while if 2^0 in R_3 is 1, this must become $+110v$. The right-hand grid of $T_{5,8}$ is driven by the left cathode of the 2C51 cathode follower $T_{4,7}$, to the grids of which is applied a voltage of $+110v$, which is caused to drop to $+15v$ when it is desired to enable the gates. Thus a 0 is signified to the Dispatch Counter by $+15v$, and a 1 by $+110v$. The circuit providing the gate command pulse to $T_{4,7}$ had not yet been installed at the time of writing.

THE GATE-CLEAR SEQUENCING CHAIN

We have seen that the Adder and Digit Resolver form the sum of the contents of R^3 (or the complement of this number) and R_1 and transmit this sum by way of a set of gates (the Green gates of RI) to R^1 . As the Green gates transmit 0's, it is necessary first to clear R^1 to 1's before opening the gates, so that it is essential that these clear and gate operations should

proceed in the proper time sequence.

The contents of R^1 can now be transferred to R_1 . We recall that the "shift down" operations shift the contents of R^1 either one place to the left or one to the right. In each case R_1 must be prepared to receive the information transmitted by R^1 by the appropriate clearing operation. Thus, for the left shift we clear R_1 to 1's and "shift down left" 0's (Green clear and Red gate), while for the right shift we clear R_1 to 0's and "shift down right" 1's.

The performance of multiplication and division are facilitated by the structure of the registers. In multiplication the multiplicand is placed in R^3 and the multiplier in R_2 , R_1 being cleared to 0. Then if the lowest order digit in R_2 (i.e., the contents of $2^{-39}R_2$) is 1, we add the multiplicand to the contents of R_1 , shift this sum one place to the right, and put it back into R_1 . We then shift the multiplier one place to the right and repeat the process if the new lowest order digit is a 1, while if it is 0, we merely shift the contents of R_1 one place to the right and do the same with that of R_2 : we are then ready for another step.

Similarly division is performed by a sequence of subtractions and shifts.

Each of these recording and shifting operations requires the following sequence: clear, gate, clear, gate: we refer to such a sequence as a "cycle" of the arithmetic unit, and to a single sequence of clear and gate as a half cycle. For convenience we record the possible operations:

Record:	Read Clear, Green Gate
Shift up:	Black Clear, Yellow Gate
Shift down left:	Green Clear, Red Gate

Shift down right: Yellow Clear, Black Gate

Obviously the first half cycle is either a "Record" or a "Shift Up", while the second is either "Shift Down Left" or "Shift Down Right".

The proper sequencing of these clearing and gating operations is the function of the "Gate-Clear Sequencing Control", of which a schematic appears in DWG No. 1287. We note that it consists of five main chassis, labelled "CT Bot.", "CG Bot.", etc. For simplicity we number these 1 to 5. Any tube will be referred to as T_{ij} , where i signifies the number of the chassis, and j the number attached to that tube in the schematic. There is also a small chassis "CX" containing two dual triodes and a dual diode. Normally the switch in the lead to cathode (1) of T_{21} is set in the position which connects this cathode to cathode (7) of T_{25} . The single pole single throw switches in the leads to cathode (7) of T_{41} and to cathode (5) of the diode on chassis "CX" are normally closed.

Inputs to the Sequencing Control are the gate and clear voltages from RI and the Counter Stop voltage, which is that labelled "Out" on the schematic (DWG No. 1289) of the Shift Counter and Recognition Circuit. This voltage has two levels: it remains at +10v until the counter contents agree with the "number to be recognized", at which time it falls to -27v. Outputs are the four voltages shown to the cathodes of the clear selector tubes and the four voltages to the gate driver-drivers in RI and RII.

Thus, it appears that the circuit generates signals which initiate gating and clearing operations, the results of which are fed back to it. One will expect, therefore, that an action once initiated will be self-perpetuating, and this is indeed the case. The Counter Stop signal serves to initiate and to terminate the process, as will be shown.

We first describe in detail the quiescent state of the circuit.

Suppose that the gate and clear inputs are all at their upper levels, that the toggles in chassis 1 and 5 have been set in the 1 position (neon lamp glowing), and that the Counter Stop voltage is at its lower level of about -2V ; which guarantees that grid (5) of T_{41} is held far below cut-off and consequently that the voltage of plate (1) is high ($+110\text{V}$).

Since the toggle in chassis 1 is in its 1 condition, the voltage of grid (5) of T_{14} is approximately 0V , so that the voltage of plate (2) is slightly less than 50V , while grid (6) is far below cut-off, and hence the voltage of plate (1) is $+110\text{V}$. The plates of T_{14} are connected to the grids of T_{16} , both sections of which are connected as cathode followers. From the cathode circuit of the left section, outputs are taken to the grid of the left section of T_{15} , and to the grid of the left section of T_{26} ; similar connections are made from the cathode circuit of the right section to the grid of the right section of T_{15} and that of the left section of T_{46} . The voltage divider network from which these outputs are taken are so designed that when the cathode voltage assumes its high value (approximately $+110\text{V}$), a slight grid current is drawn by T_{15} and the grid is held approximately at 0V , and consequently the voltage at the other end of the 1K resistor becomes approximately 7V . On the other hand, when the cathode assumes its low level of voltage, the corresponding section of the T_{16} is cut off, the grid voltage becoming approximately -35V , while that at the other end of the 1K resistor becomes -20V . Now consider the effect on T_{15} . Its grid voltages are either -35V or 0V ; hence, with a 5.6K plate resistor, the plate voltages are either $+110\text{V}$ or approximately $+60\text{V}$. The circuit in chassis 5 is exactly the same and so needs no further discussion.

The outputs of the supertoggle consisting of T_{41} and the tube mounted in the chassis "CX" are very simple: from the cathode (2) of the left section of the latter to the grid (6) of T_{34} and grid (5) of T_{35} , and from the cathode (8) of the right section to grid (6) of T_{36} and grid (5) of T_{35} . These cathodes can assume two levels: when the toggle holds a 1 the voltage of cathode (2) is approximately +60v and that of cathode (8) is +110v. We assume that the Counter Stop voltage is at its lower level of about -27v, which guarantees that the left half of the Supertoggle tube T_{41} is cut off. We shall see presently that the other section is also held off under these circumstances.

We can now summarize the approximate voltages presented to the remainder of the circuit by the tubes in chassis 1 and 5 and the supertoggle:

T_{26}	grid (3):-29v	grid (7):-29v	cathode (2):-29v	cathode (8):-29v
T_{46}	grid (3):+7v	grid (7):+7v	cathode (2):+7v	cathode (8):+7v
T_{33}	grid (5):+110v	grid (6):+60v	cathode (7):+110v	
T_{34}	grid (5):+60v	grid (6):+110v	cathode (7):+110v	
T_{35}	grid (5):(see below)	grid (6):+110v	cathode (7):+110v	
T_{36}	grid (5):+110v	grid (6):(see below)	cathode (7):+110v	

These tubes are all cathode followers; in T_{26} and T_{46} the sections are independent, but in each of the rest the cathode is common to both sections, so that the cathode voltage is in each case equal to that of the higher grid. The voltage dividers in the cathode circuits of T_{33} , ..., T_{36} are so designed that the output is +8v when one of the grids is high (+110v) and -18v when both grids are low (60v).

We now list the grid voltages of the tubes driven by the cathode followers listed above:

II	T ₂₂	grid (5) +8v	grid (6) +8v	cathode (7) +8v
	T ₂₃	grid (5) -29v	grid (6) +8v	cathode (7) +8v
	T ₂₄	grid (5) -29v	grid (6) +8v	cathode (7) +8v
	T ₂₅	grid (5) -29v	grid (6) -29v	cathode (7) -29v
	T ₄₂	grid (5) +8v	grid (6) +8v	cathode (7) +8v
	T ₄₃	grid (5) +8v	grid (6) +7v	cathode (7) +8v
	T ₄₄	grid (5) +8v	grid (6) +7v	cathode (7) +8v
	T ₄₅	grid (5) +7v	grid (6) +7v	cathode (7) +7v

We also observe that the left section (cathode (1) and plate (7)) of the diode T₂₁ is in a conducting state. This holds the grid (5) of T₄₁ below cutoff, so that grid (6) of T₃₆ and grid (5) of T₃₅ are both held at +110v.

The output voltages are now easily obtained:

III	To R ¹ Clear Selector	+8v	Red Gate	+8v
	To R ₁ Clear Selector	+8v	Black Gate	+8v
	To R ² Clear Selector	+8v	Green Gate	+8v
	To R ₂ Clear Selector	+8v	Yellow Gate	+8v

We recall that a positive input to the gate driver drivers disables the gates, while a sufficiently negative one enables them. The disabling voltage of +8v becomes more positive after cathode rises in the cathode followers which serve as gate drivers and gate driver drivers; in practice, the voltages of the -300v, +110v, and +220v buses were all dropped five percent, and the voltage dividers in the cathode circuits of the last four tubes in chassis 3 were then padded until the actual gating voltage was observed to be +10v.

Furthermore, we recall that the clear selector tubes are 6J6's whose grid voltages are the grid voltages of the appropriate toggle, and are thus

either 0v or -40v, while the cathode of the tube is directly connected to the Clear Selector output of the Sequencing Chain. With this cathode held at +8v, obviously both sections of the Clear Selector tube are cut off, and the clear bus voltage is held at +150v, while when the cathode is dropped midway between 0v and -40v, the section whose grid is at 0v conducts, while the other remains cut off.

Hence, the condition of the Sequencing Chain described above is a stable one: no clears or gates are generated and hence none are fed back to change the condition of the Sequencing Chain circuitry.

Now let the Counter Stop voltage be raised from -27v to +10v. This permits grid (6) of T_{41} to rise in voltage to 0v, beyond which it cannot go (more than a few tenths of a volt) because of the grid current that starts to flow. Thus, the supertoggle is put in the 1 condition and the voltage of cathode (2) of the supertoggle cathode follower falls to +60v, which brings the voltages of grid (6) of T_{34} and grid (5) of T_{33} both down to +60v. Referring to table I it is clear that this action causes the cathode voltages of these two tubes also to fall to +60v, and hence grids (5) of T_{42} and T_{44} , grid (6) of T_{42} , and grid (5) of T_{43} all fall from +8v to -18v. Reference to table II now shows that the cathode voltage of T_{42} falls to -18v, but those of T_{43} and T_{44} are both caught at +7v by the other grids of those tubes. Thus the net effect so far of the rise in the Counter Stop voltage is to cause the R^1 and R^2 Clear Selector voltages to fall to -18v: we need only concern ourselves with the clearing of R^1 , as it is this voltage which is fed back to the Sequencing Control.

Thus we have:

T ₂₆	grid (3):-29v	grid (7):-29v	cathode (2):-29v	cathode (8):-29v
T ₄₆	grid (3):+7v	grid (7):+7v	cathode (2):+7v	cathode (8):+7v
T ₃₃	grid (5):+60v	grid (6):+60v	cathode (7):+60v	
T ₃₄	grid (5):+60v	grid (6):+60v	cathode (7):+60v	
T ₃₅	grid (5):+110v	grid (6):+110v	cathode (7):+110v	
T ₃₆	grid (5):+110v	grid (6):+110v	cathode (7):+110v	

whence

T ₂₂	grid (5):+8v	grid (6):+8v	cathode (7):+8v
T ₂₃	grid (5):-29v	grid (6):+8v	cathode (7):+8v
T ₂₄	grid (5):-29v	grid (6):+8v	cathode (7):+8v
T ₂₅	grid (5):-29v	grid (6):-29v	cathode (7):-29v
T ₄₂	grid (5):-18v	grid (6):-18v	cathode (7):-18v
T ₄₃	grid (5):-18v	grid (6):+7v	cathode (7):+7v
T ₄₄	grid (5):-18v	grid (6):+7v	cathode (7):+7v
T ₄₅	grid (5):+7v	grid (6):+7v	cathode (7):+7v

and the outputs become:

	To R ¹ Clear Selector	-18v	Red Gate	+8v
	To R ₁ Clear Selector	+8v	Black Gate	+8v
III'	To R ² Clear Selector	-18v	Green Gate	+7v
	To R ₂ Clear Selector	+8v	Yellow Gate	+7v

Whether the Red Clear (R^1 to 1) or the Black Clear (R^1 to 0) takes place now depends upon whether $2^{-39}R_2$ holds a 1 or a 0, but obviously one of these must occur. Suppose, for example, that the Black Clear occurs, the voltage of the Black Clear bus falling from +150v to +50v. Evidently this

also clears the toggle in Chassis 1 to 0, and we must trace the consequences of this event. It is easiest to show these by another series of tables:

T ₂₆	grid (3):+7v	grid (7):-29v	cathode (2):+7v	cathode (8):-29v
T ₄₆	grid (3):+7v	grid (7):-29v	cathode (2):+7v	cathode (8):-29v
T ₃₃	grid (5):+60v	grid (6):+60v	cathode (7):+60v	
T ₃₄	grid (5):+110v	grid (6):+110v	cathode (7):+110v	
T ₃₅	grid (5):+110v	grid (6):+110v	cathode (7):+110v	
T ₃₆	grid (5):+60v	grid (6):+110v	cathode (7):+110v	

These changes in turn influence the tubes in chassis 2 and 4 as follows:

T ₂₂	grid (5):+8v	grid (6):+8v	cathode (7):+8v
T ₂₃	grid (5):+7v	grid (6):+8v	cathode (7):+8v
T ₂₄	grid (5):-29v	grid (6):+8v	cathode (7):+8v
T ₂₅	grid (5):+7v	grid (6):-29v	cathode (7):+7v
T ₄₂	grid (5):+8v	grid (6):-18v	cathode (7):+8v
T ₄₃	grid (5):-18v	grid (6):-29v	cathode (7):-18v
T ₄₄	grid (5):+8v	grid (6):+7v	cathode (7):+8v
T ₄₅	grid (5):+7v	grid (6):-29v	cathode (7):+7v

Thus we have:

III"	To R ¹ Clear Selector:	+8v	Red Gate:	+8v
	To R ₁ Clear Selector:	+8v	Black Gate:	+8v
	To R ² Clear Selector:	+8v	Green Gate:	+8v
	To R ₂ Clear Selector:	+8v	Yellow Gate:	-18v

Thus the Black Clear is terminated and the Yellow Gate enabled. Lags through the circuits cause an interval of 1.5 μ sec. to elapse between the initiation and termination of the Black Clear.

The Yellow Gate which is now enabled is fed back to the Sequence Control: the voltage of grid (6) of T₅₁ drops to about -20v and enables the other section to draw plate current: this is, of course, the standard gating procedure for reading into a toggle, and the toggle in chassis (5) is caused to flip to its 0 state. It is now necessary to trace the consequences of this, which we do again by listing the voltages throughout the circuit.

T ₂₆	grid (3):+7v	grid (7):+7v	cathode (2):+7v	cathode (8):+7v
T ₄₆	grid (3):-29v	grid (7):-29v	cathode (2):-29v	cathode (8):-29v
T ₃₃	grid (5):+60v	grid (6):+110v	cathode (7):+110v	
T ₃₄	grid (5):+110v	grid (6):+60v	cathode (7):+110v	
T ₃₅	grid (5):+110v	grid (6):+60v	cathode (7):+110v	
T ₃₆	grid (5):+60v	grid (6):+110v	cathode (7):+110v	

Thus the voltages in chassis 2 and 4 are:

T ₂₂	grid (5):+8v	grid (6):+8v	cathode (7):+8v
T ₂₃	grid (5):+7v	grid (6):+8v	cathode (7):+8v
T ₂₄	grid (5):+7v	grid (6):+8v	cathode (7):+8v
T ₂₅	grid (5):+7v	grid (6):+7v	cathode (7):+7v
T ₄₂	grid (5):+8v	grid (6):+8v	cathode (7):+8v
T ₄₃	grid (5):+8v	grid (6):-29v	cathode (7):+8v
T ₄₄	grid (5):+8v	grid (6):-29v	cathode (7):+8v
T ₄₅	grid (5):-29v	grid (6):-29v	cathode (7):-29v

And so we have

To R ¹ Clear Selector:	+8v	Red Gate:	+8v
To R ₁ Clear Selector:	+8v	Black Gate:	+8v

III"	To R ² Clear Selector:	+8v	Green Gate:	+8v
	To R ₂ Clear Selector:	+8v	Yellow Gate:	+8v

so that the Yellow Gate has been disabled. Again, due to delays in the circuitry, the disabling of the Yellow gate occurs about 1.5 μ sec. after its enabling. Another effect is that since the cathode voltage of T₄₅ falls to -29v, conduction takes place in the left section of the CX diode and in the right section of diode T₂₁, which causes the grid (6) of supertoggle tube T₄₁ to fall below cut off. As cathode (1) of T₂₁ is held at +7v, there is nothing to prevent the flipping of the supertoggle which accordingly assumes its 0 condition. This in turn raises the voltages of grid (6) of T₃₄ and grid (5) of T₃₃ to +110v, and lowers those of grid (6) of T₃₆ and grid (5) of T₃₅ to +60v: thus

T ₂₆	grid (3):+7v	grid (7):+7v	cathode (2):+7v	cathode (8):+7v
T ₄₆	grid (3):-29v	grid (7):-29v	cathode (2):-29v	cathode (8):-29v
T ₃₃	grid (5):+110v	grid (6):+110v	cathode (7):+110v	
T ₃₄	grid (5):+110v	grid (6):+110v	cathode (7):+110v	
T ₃₅	grid (5):+60v	grid (6):+60v	cathode (7):+60v	
T ₃₆	grid (5):+60v	grid (6):+60v	cathode (7):+60v	

and

II"	T ₂₂	grid (5):-18v	grid (6):-18v	cathode (7):-18v
	T ₂₃	grid (5):+7v	grid (6):-18v	cathode (7):+7v
	T ₂₄	grid (5):+7v	grid (6):-18v	cathode (7):+7v
	T ₂₅	grid (5):+7v	grid (6):+7v	cathode (7):+7v
	T ₄₂	grid (5):+8v	grid (6):+8v	cathode (7):+8v
	T ₄₃	grid (5):+8v	grid (6):-29v	cathode (7):+8v

II""	T_{44}	grid (5):+8v	grid (6):-29v	cathode (7):+8v
	T_{45}	grid (5):-29v	grid (6):-29v	cathode (7):-29v

the output voltages becoming:

III""	To R^1 Clear Selector:	+8v	Red Gate:	+7v
	To R_1 Clear Selector:	-18v	Black Gate:	+7v
	To R^2 Clear Selector:	+8v	Green Gate:	+8v
	To R_2 Clear Selector:	-18v	Yellow Gate:	+8v

Hence, a Yellow or Green Clear is made possible: which is to be performed is determined by the R_1 Clear Selector, which is controlled by the "multiply/divide" toggle. We also observe that the state of the supertoggle remains unchanged, since the cathode voltages of T_{25} and T_{45} do not change.

We have traced in detail one half cycle of operation of the sequencing chain. We began with the toggles in chassis 1 and 5 cleared to 1, the Counter Stop voltage low, and consequently both sections of the supertoggle tube cut off. This was a stable state of the circuit, which in this condition permitted no gating or clearing operations to be performed. When the Counter Stop voltage was raised to its higher value, the supertoggle assumed its 1 condition. This dropped the voltage output to the R^1 and R^2 Clear Selector, which decided, on the basis of the contents of $2^{-39}R_2$, whether to perform a Black or Red Clear. We assumed that the Black Clear was called for. The Black Clear being fed back to the Sequencing Chain flipped the toggle in chassis (1) to 0 which had two consequences: 1) the Black Clear was terminated, and 2) a Yellow Gate was enabled. Again the feeding back of the Yellow Gate signal flipped the toggle in chassis (5) to 0. This action had again two consequences: 1) the Yellow Gate was terminated, and 2) the supertoggle was

flipped to 0. This in turn permitted the performance of a Yellow or Green Clear, depending upon the state of the Multiply/Divide toggle.

If the Red Clear had been called for, the sequence would have been: flip the toggle in chassis (5) to 0, terminate the Clear, enable the Green Gate, flip the toggle in chassis (1) to 0, disable the Green Gate, flip the supertoggle to 0.

By continuing the process, we find that, providing the Yellow Clear was called for, the signal fed back to the Sequencing Chain flips the toggle in chassis (1) to 1 which terminates the Clear and enables the Black Gate. This in turn being fed back flips the toggle in chassis (5) to 1 disables the Black Gate, and flips the supertoggle again to 1. If the Green Clear is called for, the toggle in chassis (5) is first flipped to 1, the clear terminated and the Red Gate enabled. This in turn is fed back to flip the toggle in chassis (1) to 1, disable the gate, and finally to flip the supertoggle back to 1.

Thus beginning with all three toggles in the 1 condition, first a Red or Black Clear is performed followed by a Green or a Yellow Gate, respectively, at the end of which all the toggles are in the 0 condition: this is one half cycle. The second half cycle begins with a Yellow or Green Clear, depending upon the arithmetical process to be performed. These are respectively followed by Black or Red Gates, at the termination of which all the toggles are again in the 1 condition, and the circuit is ready for another cycle of operation.

THE DISPATCH COUNTER

The operation of the Williams Memory requires the periodic regeneration of the information stored therein. The practical upper bound on the time

between regenerations is about one-tenth of a second; actually it is desirable to make this interval shorter, one-thirtieth of a second being satisfactory. The CRT beams must, therefore, be directed in this interval to each of the memory locations. This requires that in the interval all possible memory addresses must be generated and presented to the deflection circuits. Each memory address is specified by a ten binary digit number, one group of five digits specifying the horizontal coordinate; the remaining five the vertical coordinate. Hence, a ten stage binary counter which starts at 0 and counts until all the stages hold 1's will generate in the process all memory locations just once.

Furthermore, using as we do a one-address code, we store orders in successive locations in the memory and use them in the order in which they are stored. There is, however, an exception to this in the case where it becomes necessary to shift control, as for example, when it is desired to repeat a sequence of instructions using new values of the numerical quantities.

It appears at first glance that the requirements of the Williams Memory which have been stated above require two counters. However, these can be combined into a single device which we call the Dispatch Counter. This is a double counter of the Adder type, consisting physically of three ranks of ten toggles each and the necessary gate, clear, and carry circuits. Each count which arrives at the input causes unity to be added to the lowest order (2^0) stage and carries to be propagated as appropriate to the higher order stages.

The three ranks of toggles are referred to as the "restore", "dispatch", and "order" toggles (T_R , T_D , T_O): the T_D and T_R toggles form the counter

which generates the memory addresses for the regeneration process, while the T_O and T_D toggles form the order counter. Leads directly out of the stages of T_D are taken to the Williams Memory Deflection Generator, while leads into T_D make it possible, when it is desired to "shift control", to replace the contents of T_D by an arbitrary number. Inputs are pulses generated in the Williams Memory Local Control: according as a Regenerate or an Action cycle is to take place, these are caused to increase the number in T_R or T_O by unity.

The schematic of the Dispatch Counter is shown in DWG No. 1336; two typical stages are drawn at the left, while the tubes at the right and in the upper left corner are gate drivers.

Consider a typical stage of the Counter. The convention here is that a toggle holds 1 when the left section conducts, as can be seen by the arrangement of the neon lamps. The tube (2C51) directly above T_D provides the gates between T_D and T_R , the left section being the B_R or "down" gate and the right section the A_R or "up" gate; the 2C51 directly below T_D provides the gates between T_D and T_O , the left section being the A_O or "down" gate and the right the B_O or "up" gate. Observe that the B_R gate guarantees that T_D will hold the same bit as T_R , while clearing T_D to 1's followed by enabling the B_O gate guarantees that T_D will hold the same bit as T_O . The operation of the A gates will be explained later. The grid voltages of T_D drive cathode followers; these are both sections of a 2C51. From the cathode of the left hand section an output is taken to the deflection gates in the Williams Memory; from each section an output is tapped down slightly below the cathode. We may assume that the values of these voltages are 0v and -40v for the two states of T_D . The 6J6 directly above T_O will be referred

to as the carry gate, while the function of the double diode will be explained below.

The "dispatch" rank of toggles operates together with either the "restore" or the "order" rank as a true adder. Suppose the number held in the "restore" rank is to be increased by unity. The first step is to clear the "dispatch" rank to 0, then to open the B_R gates. This puts into T_D the number held in T_R . To this number 1 is added automatically by introducing a permanently wired in carry to the lowest order stage. This does not change the state of the T_D toggles, but sets up the voltages on the grids of the A_R gates in such a way that when subsequently T_R is cleared to 1's and the A gates opened, T_R receives a number greater by unity than the one previously held.

As we saw in studying the Williams Memory local control, the decision as to whether the next memory cycle is to be one of action or regeneration is made during the $\bar{A}$ pulse. This decision determines whether unity is to be added to the contents of T_R or T_O . The next pulse, "C1", then performs the proper clearing operation in T_D (the "B clear"), then B is used to open the B gates. At the beginning of the next cycle, "C1 T_1 " clears T_R (or T_O as the case may be) while a pulse coinciding in time with T_D opens the A gates.

Suppose that unity is to be added to the contents of T_R : then this number is read into T_D by the B_R gates, and we wish to see how the addition of unity is effected.

In each stage of T_D we can have either a 1 or a 0 held in the toggle (the Resident Digit), and either a 1 or a 0 as the carry out. This gives us four cases as shown in the table. In each case we also show the carry from the preceeding stage, the carry which must be passed on to the next stage,

the Resident Digit (D_R), and the "new" Resident Digit (D'_R):

	C in	D_R	C out	D'_R
I	0	0	0	0
II	0	1	0	1
III	1	0	0	1
IV	1	1	1	0

As we said before, the "carry" input to the lowest order stage is permanently wired to a point of negative voltage (-300v), which is the means used to introduce a 1 into the first stage in the form of an artificial carry. A carry of 1 is always given by a negative voltage, while a 0 carry is signified by a positive voltage. In the actual circuit, the lowest observed 0 carry voltage was +5v, the highest +26v, while the 1 carry voltages ranged from -23v to -27v.

We will now consider in detail each of the four cases for an arbitrary stage: of course, only cases III and IV can occur in the first stage.

I. The voltage of the right grid of the carry gate is -40v, while that of the left grid is positive. The cathode, therefore, follows the left grid high enough to cut off the right section, causing a positive (0) carry to be transmitted to the next stage. If all the resistors in the plate network of the right section of the carry gate had exactly their nominal values, this carry voltage would be +21.4v; the effect of tolerances is such that values of from +5v to +26v were observed in the ten stages of the counter, as noted above. The grid of the A_R gate would go far negative, but it is held up by conduction through the left section of the dual diode to about 0v; measured values of from 0v to +1v were observed.

Next, the T_R toggle is cleared to 1, and subsequently the A_R gate is opened by dropping the cathode voltage to -10v. As the grid voltage of the A_R gate tube is 0v, conduction takes place which flips the T_R toggle back to 0. Hence, $D_R' = D_R = 0$ and $C_{out} = 0$.

II. Though here the right grid of the carry gate attains a voltage of 0v, the cathode is still held up sufficiently by the positive voltage of the left grid that the right section is cut off. Again the carry to the next stage is a positive voltage. The left plate of the dual diode is now connected to a point below ground, while the cathode of the right section is positive. Hence, the diode is inoperative, and the grid of the A_R gate becomes quite negative -- from -28v to -31v in the ten stages of the counter.

This suffices to prevent the enabling signal to the A_R gate from having any effect, and, therefore, the T_R toggle, once cleared to 1, will remain in that condition. Hence, $D_R' = 1$ and $C_{out} = 0$.

III. Both grids of the carry gate are negative. We have mentioned that the highest value of voltage observed for a carry of 1 is -23v. The right grid of the T_D toggle is high (0v) which permits both diode sections to conduct, and thus to hold the carry gate cathode voltage so high that both sections are cut off. Hence, we obtain a positive carry voltage to the next stage, while the A_R gate grid voltage is observed to lie between -10v and -12v.

Thus when the A_R gate is enabled, it does not draw sufficient current to flip the T_R toggle from 1, to which it has been cleared, back to 0. Hence, $D_R' = 1$, $C_{out} = 0$.

IV. Here the cathode of the carry gate follows the right grid high enough to cut off the left section, giving a negative (1) carry to the next

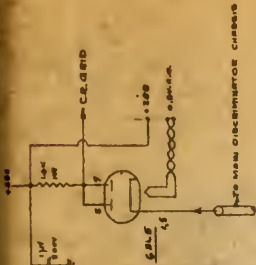
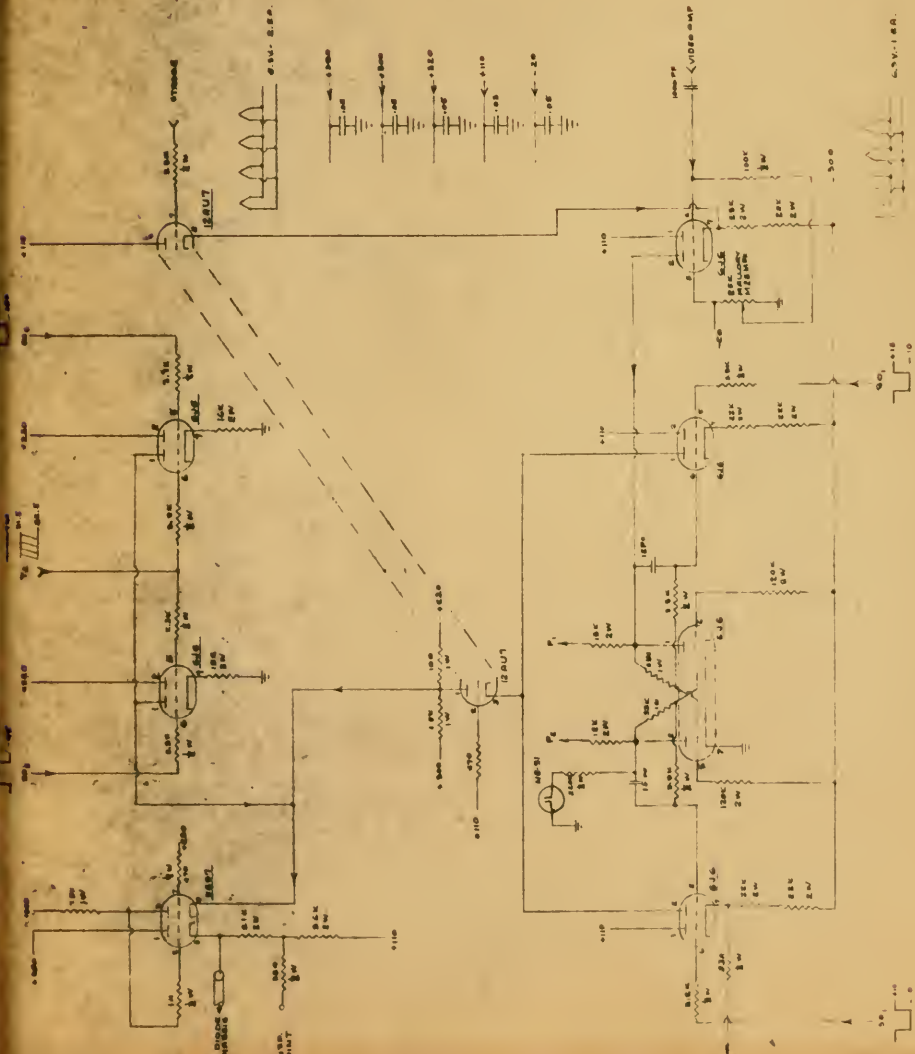
stage. The grid of the A_R gate is prevented from going positive by conduction through the right section of the diode: observed values of grid voltage range from 0v to -1.5v.

Hence when the A_R gate is enabled, sufficient current is drawn to flip T_R back to 0, and we have $D_R' = 0$, C out = 1.

Thus, in the two cases (II and III) in which the "new" digit in the stage under consideration, resulting from the addition of the carry from the previous stage to the resident digit, is a 1, the A_R gate grid voltage is always less than -10v, while in the two cases (I and IV) in which it is a 0, the A_R gate grid voltage lies in the range from -1.5v to +1v. Thus we see why it is necessary to clear T_R to 1's before opening the A_R gates; furthermore, it is clear that the A_R gates are kept closed as long as the cathode voltage is +10v, which is a convenient level obtained from the Williams Memory Local Control. The opening of the gates can be accomplished by dropping the cathode below the +1v, -1.5v range. Faster action is obtained the lower the cathode is dropped, but it is necessary to keep this level sufficiently high that a grid voltage of -10v will not result in the drawing of sufficient current to flip the T_R toggle: a lower value of -6v instead of -10v would appear to be reasonable here.

This suffices to illustrate the operation of the Dispatch Counter. A complete account of its functioning in the system must await the completion of the Main Control.

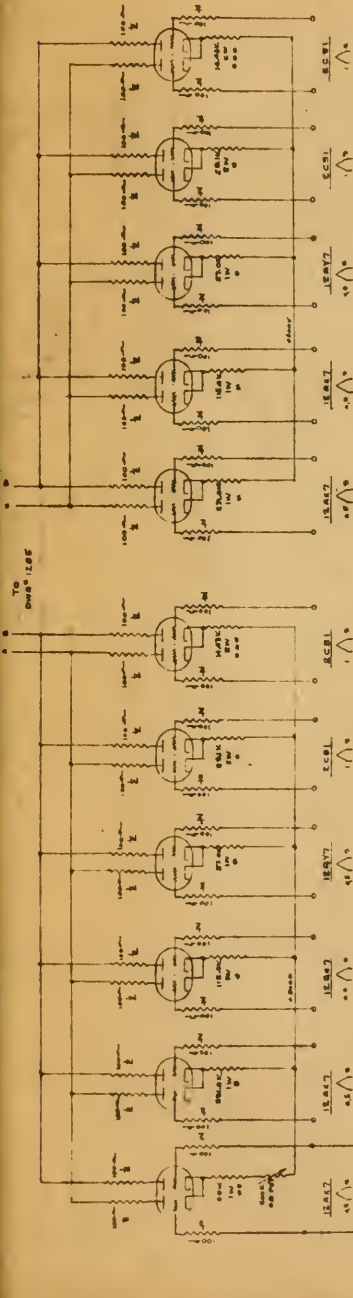




BIORG, CHARLES EUGENIUS
(BIRTH RECORD NUMBER 1090)

DC VOLTS	PULSE	HERTZES
+200	501	535-100 C.V.D.C.
+500	504	535-220
+1500	502	
	502	INPUT - AMP.
	502	OUTPUTS - C.F.G.R.D.
	502	9-AVE
	502	P. (-100)
	500	P. (100)

1000000
1000000
1000000



ADVERSE DEFLECTION RATE
LEVEL ADJUSTMENT

TUBES	WATER LEVEL
6X4	0.000
6AR5	0.010
6AV6	0.020
6X4	0.030
6AR5	0.040
6AV6	0.050

THIS CIRCUIT REPEATED FOR EACH OF ABOVE TUBES



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DEFLECTION INPUT &
ORDER SYSTEM

DRAWN BY: DATE: 1-27-50
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One square of 10, 36 + 36 + 72

FOR 5 GUY PERSONS NEW CAPITAL

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DWG. No. 1233

